

Artifact-Gated Evaluation of a Causal FIR Module for Sequential Recommendation

[MAINTAINER: REAL AUTHOR NAME(S) REQUIRED – TORS IS SINGLE-BLIND], [Maintainer: real institution required – TORS is single-blind], [Maintainer: country]

We evaluate a small, detachable causal finite-impulse-response (FIR) residual for sequential recommendation. Outcome-known Amazon Reviews 2023 estimates are positive on Musical_Instruments (+0.002265, ordinary Welch 95% CI [0.001928, 0.002602]), Industrial_and_Scientific (+0.002110, ordinary paired CI [0.001820, 0.002399]), and CDs_and_Vinyl (+0.006150 [0.005849, 0.006450]). A frozen Software attempt found learned FIR minus identity +0.005062 [+0.004591, +0.005533], but unverifiable prior-output non-visibility keeps it outcome-known same-team robustness rather than independent confirmation. On Musical_Instruments, learned FIR exceeded an equal-parameter current-position-only DCT/GELU placebo by +0.001941 [+0.001788, +0.002095], yet did not separate from a shared causal filter (-0.000081 [-0.000337, +0.000175]). Most importantly, a prospectively frozen same-investigator MovieLens 1M study returned ML1M-NO-FIR-REPLICATION: learned FIR minus identity was +0.000000 [-0.000074, +0.000075] ($p_{\text{Holm}} = .995$) and learned FIR minus pointwise was +0.000035 [-0.000057, +0.000127] ($p_{\text{Holm}} = .796$). Shared (16 parameters), grouped (128), and low-rank (320) FIR arms met the pre-declared 0.000500 noninferiority margin versus the 1,024-parameter learned FIR, but this is only a conditional compression result because the learned-FIR replication gate failed. Two frozen current-comparator studies under our evaluator bracket the package: an AlphaFuse-style MiniLM representation package scored 0.04827 [0.04813, 0.04842] versus its repository SASRec-ID backbone at 0.03902 [0.03811, 0.03994], while WEARec scored 0.05918 [0.05867, 0.05969]; both remained below the existing 0.06734 full-model reference. These are outcome-known same-investigator whole-package/equal-evaluation results, not causal isolation, equal capacity or tuning, independent confirmation, or SOTA. Thus the evidence supports a narrow modular implementation and positive internal Amazon contrasts, not a general FIR benefit, per-channel necessity, or cross-domain generalization. A fail-closed graph preserves positive, null, deviated, and VOID outcomes under the same reporting rule.

CCS Concepts: • **Information systems** → **Recommender systems**; *Personalization*.

Additional Key Words and Phrases: sequential recommendation, Amazon Reviews 2023, HSTU, causal FIR filtering, long-tail recommendation, text-augmented recommendation, reproducibility, pre-declaration

1 Introduction

Sequential recommendation models predict a user’s next item from their interaction history. SASRec [Kang and McAuley 2018] and BERT4Rec [Sun et al. 2019] established Transformer-based baselines. Later systems add language-derived item representations [Ding et al. 2021; Hou et al. 2024, 2022; Li et al. 2023] or discrete semantic IDs [Rajput et al. 2023; Yang et al. 2024].

Amazon Reviews 2023 results span incompatible preprocessing variants. In particular, the 0-core reference pipeline and the 5-core protocol used by recent generative retrievers do not define interchangeable benchmarks. Our experiments use the latter: fixed 5-core splits, leave-last-two-out histories, and full-catalog ranking.

Recommender evaluation is vulnerable to mistuned baselines, incompatible protocols, and irreproducible result selection [Ferrari Dacrema et al. 2021, 2019]. We therefore pair the method with a per-paper discipline: git-frozen protocols, a fail-closed graph that recomputes 199 reported cells across 23 claim families, environment-caveated comparator

Author’s Contact Information: [Maintainer: real author name(s) required – TORS is single-blind], [Maintainer: real institution required – TORS is single-blind], [Maintainer: country].

regeneration, and symmetric adjudication. Failures are retained. In particular, the original Office_Products protocol remains VOID after its own floor check exposed an environment mismatch, although a redesigned fresh-seed protocol later passed (§5.2; Appendix A.0).

Using an HSTU-style pure-PyTorch encoder based on Zhai et al. [2024], we ask whether one small causal filtering component survives matched controls and can be reused without category-specific tuning. Our contribution ledger is deliberately limited to three items:

- (1) **A canonical causal FIR module.** We adapt earlier bidirectional frequency filters to a strictly left-causal, depth-wise residual that is identity-initialized and gradient-active under the all-position objective (§3). This is a modular component, not a new recommender architecture.
- (2) **Internal evidence, a negative non-Amazon test, and a mechanism boundary.** Learned FIR–identity estimates are positive on three outcome-known Amazon category/split settings. A separately frozen Soft-ware attempt then produced +0.005062 [+0.004591, +0.005533], clearing its pre-declared +0.000500 reporting threshold with validation-only selection and sealed final evaluation. Because earlier V2 validation-output non-visibility is not independently established, V3 is classified outcome-known/exploratory under local same-user custody—not confirmation. On Musical_Instruments, learned taps do not separate from a shared causal filter, but they beat a parameter-matched current-position-only placebo by +0.001941 [+0.001788, +0.002095], while the placebo–identity interval spans zero. This discriminates learned FIR from that compound placebo; it does not isolate temporal access, establish unique necessity of per-channel taps or provide independent confirmation. A prospectively frozen MovieLens 1M study then failed to replicate learned FIR against identity or pointwise. Its shared/grouped/low-rank arms met a pre-declared noninferiority margin versus learned FIR only conditionally on that failed replication gate. Together these results prevent a general filtering or cross-domain claim. An official-code WEARec feasibility run under the same evaluator also scored below the existing full-model reference; it equalizes evaluation only, not architecture, loss, schedule, or tuning, and is neither independent confirmation nor SOTA evidence. A frozen AlphaFuse-style package-versus-repository-ID study returned a positive whole-package contrast but remained below that full-model reference; it improves comparator coverage without isolating a component or supplying independent confirmation.
- (3) **An auditable evaluation record.** Released protocols, artifacts, sidecars where available, comparator-regeneration code, and a strict rebuild preserve positive, null, deviated, and VOID outcomes under the same reporting rule (§8). The apparatus certifies reconstruction and protocol history; it does not strengthen the estimand.

TAPE, late fusion, sparse-warm redistribution, titrations, probe screens, and earlier Beauty results are supporting studies. All borrowed architectures and training recipes are attributed in §2.

2 Related Work and Attribution

We summarize prior work that our experiments directly build on. A complete attribution table for every component used in our experiments appears at the end of this section.

SASRec [Kang and McAuley 2018] is the base Transformer sequential recommender we re-implement. SASRec models a user’s history with a causal-masked Transformer encoder and scores the next item via dot product with a learned item-embedding table.

BERT4Rec [Sun et al. 2019] adapts BERT-style bidirectional masked-item modeling for sequential recommendation. We implement BERT4Rec as one of our 20 ablation baselines on Beauty_and_PC.

SBERT / MiniLM [Reimers and Gurevych 2019; Wang et al. 2020] provide pre-trained sentence embeddings. We use sentence-transformers/all-MiniLM-L6-v2 [Sentence-Transformers maintainers 2021] (384-d) as the default frozen item-text encoder.

UniSRec [Hou et al. 2022] is an early, widely-adopted text-based sequential recommender that uses a pre-trained language model to embed item text and learns a parametric adaptor to project these embeddings into the sequential model’s hidden space; **ZESRec** [Ding et al. 2021] precedes it in representing items from natural-language descriptions with a pre-trained encoder for zero-shot sequential recommendation, and **RecFormer** [Li et al. 2023] subsequently studies language representations for sequential recommendation including low-resource and cold-start evaluation — we make no priority claim in this line. The SASRec-with-frozen-text-encoder pattern we use is a strict simplification of UniSRec.

BLaIR [Hou et al. 2024] (published as Hou et al. 2026b) is a RoBERTa-base model continually pretrained on 10% of (item-metadata, review) pairs from Amazon Reviews 2023. We use hyp1231/blair-roberta-base (768-d) as an alternative frozen item-text encoder. Hou et al. also publish a reference implementation, **SASRecText**, that pairs SASRec with BLaIR via a 2-layer MLP adaptor [768, 300, 64] with Dropout(0.2) + ReLU; we adopt this adaptor design verbatim (see Appendix A.1).

TIGER [Rajput et al. 2023] reformulates sequential recommendation as autoregressive generation over discrete semantic IDs produced by a residual-quantized variational autoencoder over item-text embeddings. TIGER reports strong results on Amazon Reviews 5-core benchmarks.

LIGER [Yang et al. 2024] extends TIGER with a dense-retrieval refinement step and reports further gains.

Amazon Reviews 2023 [Hou et al. 2024] is the source dataset. We use the **5-core leave-last-out** protocol (described in §3) matching the convention used by TIGER, LIGER, and similar works, which differs from the 0core_timestamp_w_his_* HuggingFace subset used by Hou et al.’s seq_rec_results/ reference implementation.

Evaluation practice and reproducibility in recommender systems. Our lead contribution responds to a documented evaluation-trust problem in this field. Ferrari Dacrema et al. [2019] examined 18 recent neural recommenders and could reproduce only 7 with reasonable effort, 6 of which were often outperformed by comparably simple heuristic baselines; the extended journal analysis [Ferrari Dacrema et al. 2021] corroborates the pattern in greater depth and attributes it in part to the choice and optimization of the baselines used for comparison. Sun et al. [2020] respond at the community level with standardized benchmarking for reproducible evaluation and fair comparison. These works diagnose the problem and propose field-level remedies; what this paper adds is complementary and *per-paper*: an executable discipline that a single submission can adopt and a reviewer can re-run — version-controlled pre-declaration of the comparator-facing confirmation (§5.2), a fail-closed artifact gate that recomputes every artifact-gated table cell from released artifacts at build time (published comparator values are pinned as labeled external constants; §2.2 records a retirement the gate forced), local regeneration of the comparator by its own unmodified reference implementation (environment-caveated single-run regenerations, §5.6), and symmetric publication of failures, including a pre-declaration VOIDed by its own comparability check (Appendix A.0) with a redesigned successor that subsequently passed on fresh seeds (PREREG_OFFICE_V3.md; §5.2), the original VOID standing unchanged. None of this apparatus strengthens any empirical claim — every non-claim of §5–§7 stands unchanged; it certifies that the numbers printed are the numbers the artifacts recompute, under decision rules frozen before the confirmatory runs.

2.1 Attribution table

Component	Source	Where in our work
SASRec architecture	Kang and McAuley [2018]	base SASRecSBERT model
BERT4Rec	Sun et al. [2019]	one of 20 ablation baselines
MiniLM all-MiniLM-L6-v2 (384-d)	Reimers and Gurevych [2019]; Wang et al. [2020]	default frozen item-text encoder
BLaIR blair-roberta-base (768-d)	Hou et al. [2024] (arXiv:2403.03952)	alternative frozen item-text encoder
2-layer MLP adaptor [768, 300, 64] Dropout+ReLU	Hou et al. [2024] (SASRecText, in external/AmazonReviews2023/seq_rec_results/)	--mlp-adaptor flag
Rich-text content concatenation (title + categories + store name)	derived from Hou et al. [2024] preprocessing	encode_richtext_5core.py
Amazon Reviews 2023 dataset	Hou et al. [2024]	all benchmarks
TIGER / LIGER	Rajput et al. [2023] / Yang et al. [2024]	published comparator numbers

2.2 Our additions to the above prior art

We do not claim a wholly new recommender architecture. Our architectural additions are deliberately small: a causal FIR adaptation of prior frequency-filter ideas and a soft text-prototype additive term. Our contributions are:

- An **open-source 5-core preprocessing pipeline** (`preprocess_5core_standard.py`) that matches the protocol used by TIGER and LIGER (not the 0-core protocol used by Hou et al.’s repo).
- A **memory-efficient chunked-full-softmax loss** that enables proper full-catalog cross-entropy training at 207k-item catalogs on a single 16 GB GPU.
- An **eval optimization** (caching `all_item_features()` once per `evaluate()` call) that gives $\sim 10\times$ speedup at $d=64$ with the MLP adaptor.
- A **cross-pipeline transfer study**: empirically testing which published architectural choices help on the 5-core protocol.
- **Negative-result documentation** for 18 of 20 tested variants on Beauty_and_Personal_Care 5-core.
- The **competitive Video_Games 5-core result** on the HSTU-style pure-PyTorch stack (6-seed NDCG@10 = **0.0673 $\pm$ 0.0003**, +17.5% over published SASRec 0.0573), shown to be architectural (ID-only ≈ 0.0656), with no SOTA claim. (*The earlier v1-era SASRec-SBERT number 0.05509 $\pm$ 0.00035 has been retired from Table 1a: its raw per-seed artifacts predate the artifact manifest and cannot be recomputed, so it survives only as a RETIRED provenance row in the manifest, not as a paper claim.*)
- The two added components — the **strictly causal FIR temporal filter** (positive matched identity contrasts on three outcome-visible categories plus a positive outcome-known Software robustness contrast; learned FIR separates from one equal-parameter compound non-temporal placebo in MI, while the shared causal control prevents

per-channel-tap attribution; §5.2) and **TAPE** — plus the **MI frequency-5 tail case** (§5.3) and its **interaction-thinning titration** level-contrast result (§5.4). The novelty boundary for each is stated explicitly in Table 0 (§2.3).

2.3 Claim boundary

Table 0 separates prior work from the paper’s actual claim boundary. Frequency filtering (including FEARec, FreqRec, WEARec, and WPGRec) and causal convolution (including Caser and NextItNet) predate this work [Du et al. 2023; He et al. 2026; Liu et al. 2026; Tang and Wang 2018; Xu et al. 2026; Yuan et al. 2019], as do residual identity initialization, HSTU, frozen text encoders, and label smoothing. Our claim is limited to the gradient-active left-causal depthwise-FIR residual used before an HSTU-style stack under an all-position objective, together with its matched internal evaluation. The historical zero-gated realization is singular at initialization and remains package-attributed; the nonsingular canonical realization permits a FIR-arm interpretation, while the learned FIR separates from one equal-parameter current-position-only DCT/GELU placebo, but that compound contrast does not isolate temporal access; a shared causal filter prevents attribution to per-channel taps (§3, §5.2).

Table 0: Claim boundary — reused basis, specific change, and supported evidence.

Component and reused basis	Specific change	Evidence and boundary
HSTU base — Zhai et al. (2024): pointwise $\text{silu}(\text{QK}^\top + \text{rab}) \vee$, per-block normalization, and relative biases	Pure-PyTorch implementation; two spurious normalizations removed	Core-block parity only at a mirrored aligned configuration; no pinned end-to-end reproduction (§3.2, §5.6, §6.3). Prior-work implementation.
BLaIR/SBERT text — Hou et al. (2024), Reimers & Gurevych (2019), Wang et al. (2020)	Frozen off-the-shelf text features and Hou et al.’s MLP adaptor; no method change	Table 1a encoder comparison. Prior work.
AlphaFuse-style text+ID package — Hu et al. (2025)	Official upstream AlphaFuse/SASRec classes under one frozen configuration, with MiniLM-384 substituted for the published text vectors and the paper’s complete-history-masked evaluator	AlphaFuse-style NDCG@10 0.048273 [0.048129, 0.048416] versus repository SASRec-ID 0.039024; delta +0.009249 [+0.008329, +0.010169], but -0.019065 versus the existing 0.067337 reference. Countable outcome-known same-investigator whole-package evidence; not a published-table reproduction, null-space-fusion isolation, equal architecture/capacity/tuning, independent confirmation, or SOTA.
Label smoothing/time bias — Szegedy et al. (2016); TiSASRec [Li et al. 2020]; HSTU [Zhai et al. 2024]	Integrated with this backbone and full-catalog chunked softmax	+0.0013 and +0.0027 single flags (Table 1). Prior work.

Component and reused basis	Specific change	Evidence and boundary
Sequence/frequency filters – FMLP-Rec, BSARec, FreqRec, WEARec; causal convolutions include Caser, NextItNet, C3SASR, and AdaMCT	The official WEARec model/training code is evaluated under our split, complete-history mask, full-catalog evaluator, cutoff, and tie rule; the others are not inserted as-is	WEARec official-code/equal-evaluation mean 0.059184 [0.058674,0.059693] versus reference 0.067337; WEAREC-BELOW-EXISTING-REFERENCE. Outcome-known same-investigator feasibility baseline, not equal architecture/loss/schedule/tuning, independent confirmation, or SOTA. Filtering, frequency modeling, and causal convolution are prior art.
Causal FIR adaptation (ours) – the filtering/convolution line above	Minimal leak-free, gradient-active, identity-initialized left-causal depthwise residual before an HSTU-style all-position stack	Outcome-known FIR–identity estimates: MI +0.002265, IS +0.002110, CDs +0.006150. Software robustness: +0.005062 [+0.004591,+0.005533]. Learned beats the equal-parameter current-only placebo by +0.001941 but does not separate from shared FIR (-0.000081 [-0.000337,+0.000175]). Prospective MovieLens: learned minus identity +0.000000 [-0.000074,+0.000075] and learned minus pointwise +0.000035 [-0.000057,+0.000127]; neither gate passed, so parsimonious-arm NI is conditional only. Incremental modular contribution; no general FIR benefit or independent confirmation.
TAPE (ours) – TIGER, VQ-Rec, ProtoMF; prototype/semantic-ID motivation	Frozen soft assignments gate a zero-initialized additive prototype table	+0.0009 single flag; +0.0004 four-seed check. Supporting ablation (§5.1).
Tail/thinning analysis (ours) – standard popularity strata, MELT, DropoutNet [Volkovs et al. 2017], CLCRec [Wei et al. 2021]	Per-dataset text–ID tail contrast and matched-R1 interaction/user thinning	MI +0.000420 (outcome-visible); VG null; cross-dataset p=.13; user-mode p=.058. Secondary empirical boundary (§5.3–§5.4).

Closest filtering systems. FMLP-Rec and BSARec establish learnable sequence filtering and frequency-rescaled attention; BSARec’s Theorem 1 concerns repeatedly applied softmax attention, not our HSTU-style operator. C3SASR and AdaMCT already combine causal/local convolution with attention [Chen et al. 2022; Jiang et al. 2023]. Other nearby filtering systems include DLFS-Rec, DIFF, TASIF, TV-Rec, and ConvFormer [Kim et al. 2025; Liu et al. 2023a; Luo et al. 2026; Shin et al. 2025; Wang et al. 2023]; current frequency systems FreqRec and WEARec add inter-session spectral paths or adaptive wavelet filtering [He et al. 2026; Xu et al. 2026]. Our distinction is therefore only the small gradient-active, identity-initialized **left-causal depthwise FIR residual** under one HSTU-style all-position, full-catalog protocol—not convolution, filtering, frequency modeling, or temporal specificity as general contributions.

Closest text/ID systems. AlphaFuse [Hu et al. 2025] learns ID embeddings in the null space of language embeddings and is the closest frozen-text-plus-ID comparator. The first AlphaFuse-style MiniLM port on our Video_Games

split used mismatched seen-item masking; its repaired V2 factorial remained outcome-visible, capacity/initialization-confounded, incompletely masked for long histories, and non-rank-reconstructive, so its governed adjudicator permanently marks it NONCOUNTABLE. Separately frozen V3 used eight fresh seeds per arm, complete-history-masked validation selection, and sealed TEST evaluation. Its first-reader verdict, EEV3-REPORTABLE-OUTCOME-KNOWN, records AlphaFuse-style MiniLM NDCG@10 0.048273 [0.048129, 0.048416] versus 0.039024 [0.038106, 0.039941] for the repository SASRec-ID backbone: descriptive independent-arm difference +0.009249 [+0.008329, +0.010169]. The package nevertheless remains -0.019065 [-0.019347, -0.018783] below our existing 0.067337 reference. This counts as current-comparator evidence under shared data/evaluation and a frozen training configuration, not a reproduction of AlphaFuse’s published text encoder, null-space-fusion isolation, equal architecture/capacity/tuning, independent confirmation, or SOTA. Whitening, channel separation, prototype/codebook, decoupled fusion, and adaptive semantic/ID methods remain active prior lines [He et al. 2025; Hu et al. 2026; Lee et al. 2026; Li et al. 2025; Liu et al. 2021, 2024; Wei et al. 2026; Xie et al. 2022; Xu et al. 2024; Zhang et al. 2024, 2026; Zhou et al. 2026]. Our text findings remain construction-specific; item-text permutation remains open. Mecos and RecGPT address different few-interaction or zero-shot/short-history settings [Jiang et al. 2025; Zheng et al. 2021].

Two concurrent 2026 preprints reinforce that this is an active modular-content line rather than a priority claim: SISA-Rec injects frozen BERT semantics through gated input fusion and a semantic attention term [Abbasi et al. 2026], while ASER adds review-distilled sensory representations to four existing sequential backbones [Yoon et al. 2026]. Their Amazon Reviews 2014 protocols and representation interventions are not numerically interchangeable with our AR2023 FIR study; we cite them for mechanism-family coverage only.

Tail and reproducibility boundary. SimRec, MELT, DropoutNet, CLCRec, and LLM-ESR already address sparse, long-tail, or content-supported recommendation [Brody and Lagziel 2024; Kim et al. 2023; Liu et al. 2024; Volkovs et al. 2017; Wei et al. 2021], so our tail contribution is a measured, dataset-specific frequency-5 case rather than a method claim. LLM2Rec uses Amazon Reviews 2023 with 5-core leave-one-out/full ranking, but its length-10 histories and resulting item universe prevent direct numeric comparison [He et al. 2025]. Elliot, DaisyRec 2.0, accountability workflows, and reproduction audits predate our apparatus [Anelli et al. 2021; Bellogin and Said 2021; Petrov and Macdonald 2022; Sun et al. 2022]; we claim only this paper’s combination of per-cell recomputation, provenance tripwires, and symmetric self-VOIDing, not evaluation infrastructure generally. Chunked exact cross-entropy is likewise implementation engineering, not a contribution [Wijmans et al. 2025].

3 Method

3.1 Preprocessing pipeline

We preprocess Amazon Reviews 2023 [Hou et al. 2024] into 5-core leave-last-out splits. For each category we:

1. Load raw user-item review interactions from the raw_review_<category> HuggingFace subset.
2. Apply a 5-core filter iteratively until no user or item has fewer than 5 interactions.
3. Sort each user’s interactions chronologically.
4. Hold out the last interaction as test, the second-to-last as validation, the rest as training.
5. Encode item metadata (meta_<category>.jsonl) with the chosen frozen text encoder.

This protocol matches TIGER [Rajput et al. 2023] and LIGER [Yang et al. 2024] but differs from Hou et al. [2024]’s reference repo (external/AmazonReviews2023/seq_rec_results/) which uses the 0core_timestamp_w_his_* HuggingFace subset. The 5-core protocol produces smaller, denser interaction matrices.

Estimand and construct boundary. Each review event is treated as an implicit positive interaction regardless of its 1–5-star rating or verified-purchase flag. The estimand is therefore *ranking the next recorded review event among the fixed post-filter catalog*, not predicting preference, satisfaction, purchase, or deployment engagement. Five-core eligibility, item indices, and the candidate catalog are computed from complete histories before splitting, so future events can affect inclusion even though their identities are not used as training targets. Rating-threshold, verified-purchase, genuine implicit-event, and global-time/query-time-catalog sensitivities have not been run; conclusions do not extend to those constructs or deployment regimes.

Pre-core deduplication (disclosed 2026-07-19). Before k-core filtering, duplicate (user, item) interactions are collapsed to a single event, keeping the **earliest** timestamped occurrence (first-exposure convention). This step is material and was previously undocumented in the paper: it removes 41,888 of 3,017,439 raw rows on Musical_Instruments (1.39%), 69,115 of 4,624,615 on Video_Games (1.49%), 320,078 of 23,911,390 on Beauty_and_PC (1.34%), and 156,363 of 12,845,712 on Office_Products (1.22%); the FIR-breadth categories were preprocessed from AR2023 rating-only exports whose (user, item) pairs are already unique (0 rows removed on Industrial_and_Scientific and CDs_and_Vinyl). Deduplication can change k-core membership, sequence order, and held-out targets; sensitivity to a keep-latest rule is untested (a disclosed limitation). (The MI/Office counts are from the retained preprocessing logs; the VG/Beauty logs were not retained, and their counts were regenerated 2026-07-19 from the raw dumps with the released deterministic script.)

Timestamp-tie policies (disclosed 2026-07-19; two distinct deterministic rules). Two tie-breaking rules operate at different stages and were previously undocumented. (i) Preprocessing stable-sorts each user’s interactions by timestamp only, so for equal-timestamp events the raw-file order decides both which duplicate survives deduplication and which event lands on a leave-two-out split boundary. (ii) The trainer later re-sorts each user’s train sequence by (timestamp, item_index), so within-train equal-timestamp order is decided by reindexed item ID. Both rules are deterministic given the released raw dumps and code. Measured scale (2026-07-19, from the released splits): on Musical_Instruments, 3 of 57,439 users (0.01%) have test-target timestamp equal to the validation-target timestamp, 3 have validation equal to the last train timestamp, and 12 (user, timestamp) groups cover 24 of 511,836 rows; on Video_Games the counts are 4 and 8 of 94,762 users and 61 groups covering 122 of 814,586 rows (0.01%). At this scale the tie policies cannot materially affect any reported number; we nevertheless document them as part of the exact preprocessing contract.

3.2 Model architecture

Shared item-feature construction (both encoder families). For each item i :

- A learned per-item embedding $e_i \in \mathbb{R}^d$ (initialized $N(0, 0.02)$, padding-aware).
- A frozen text embedding $s_i \in \mathbb{R}^k$ from MiniLM ($k=384$) or BLaIR ($k=768$).
- A learnable projection $\Phi : \mathbb{R}^k \rightarrow \mathbb{R}^d$.

The item feature passed into the encoder is $f_i = e_i + \Phi(s_i)$. Sequences are right-padded with a dedicated pad token and processed with a causal-only attention mask (no key-padding mask — see §3.4). The output logits for position t are $h_t \cdot \text{all_items.T}$ where `all_items` is the (`n_items`, `d`) item-features table.

Headline encoder — HSTU-style (every §5 result). All headline experiments use the **HSTU-style pure-PyTorch pointwise-attention encoder**, described in full in §3.7: pointwise $\text{silu}(QK^\top + \text{rab})V$ aggregation with the published per-block normalization — no softmax, no $1/\sqrt{d}$ scaling — with the core block verified exactly (max

abs diff 0.0) against the reference research implementation at a mirrored, aligned, dropout-free configuration (HSTU_PARITY_REPORT.md discloses the alignment; the trained configuration is a strict-superset HSTU-style variant — §6.3). Configuration, stated here and in §4.4: **4 layers, 2 heads, d = 64, dropout 0.5**.

Baseline encoder — SASRec-SBERT (protocol parity and the superseded Appendix A.1 scans only). The SASRec-family baseline follows Kang and McAuley [2018]: a 2-layer pre-norm Transformer encoder (d=64, n_heads=2, FFN dim=4d, GELU activation) followed by a final LayerNorm, using the same item-feature construction. **No headline number uses this encoder**; it appears as the per-category parity/floor baseline and in the superseded Beauty scan (Appendix A.1).

3.3 Choice of projection Φ

Following our cross-pipeline transfer study (Appendix A.1), we compare two designs for Φ :

- **Linear:** $\Phi(s) = Ws$ for a learned weight $W \in \mathbb{R}^{d \times k}$.
- **MLP** (adopted from Hou et al. [2024] SASRecText): $\Phi(s) = W_2 \cdot \text{ReLU}(W_1 \cdot \text{Dropout}(s))$ with hidden dim 300 and Dropout(0.2) between and before the linear layers. **This design is from Hou et al. [2024] and we do not claim novelty for it.**

3.4 Right-padding + causal-only mask

During exploration we identified a numerical failure mode in the standard SASRec implementation pattern: combining left-padding with both a key-padding mask and a pre-norm Transformer encoder produces NaN logits when an entire row of keys is padded (the softmax denominator is zero). The NaN propagates through residual connections and silently inflates eval NDCG to ~0.9955 — a “fake-perfect” result. We adopt right-padding + causal-only mask (no key-padding mask) to avoid this trap.

3.5 Chunked-full-softmax loss

For training with cross-entropy over the full catalog (no sampled negatives), the full logits tensor (B, L, n_items) is prohibitive at L=50 and n_items=207k. We implement an incremental-logsumexp loss:

```
def chunked_full_softmax_loss(hidden, targets, model, item_chunk):
    lse = -inf # running logsumexp over the item dimension
    target_scores = None
    for chunk_start in range(0, n_items, item_chunk):
        chunk_emb = model.item_features(chunk_ids)
        chunk_logits = hidden @ chunk_emb.T
        lse = logsumexp(lse, logsumexp(chunk_logits, axis=-1))
        if any(target in chunk):
            target_scores[in_chunk] = chunk_logits[in_chunk, target]
    return -mean(target_scores - lse)
```

This bounds peak memory by (batch, item_chunk) instead of (batch, n_items) and allows proper full-softmax training at the 207k-item Beauty_and_Personal_Care scale on a 16 GB GPU.

Memory-efficient exact cross-entropy has published precedent (Cut Cross-Entropy, ICLR 2025, computes exact CE without materializing the full logit matrix); our chunked variant is independent implementation engineering in the same spirit, claimed only as engineering.

3.6 Evaluation protocol

We score each user against the **full catalog** of 207k items (Beauty_and_PC) or 25k items (Video_Games). For the held-out test item, we compute the rank under dot-product scoring (after masking train+val items the user has already seen) and report NDCG@10, HR@10, MRR. We use the standard SASRec test-time convention of including the validation item in the input sequence [Kang and McAuley 2018].

For our default 16 GB GPU, the optimized evaluation caches the (n_items, d) item-features table once per evaluate() call rather than recomputing it per batch — this is critical with the MLP adaptor, which would otherwise recompute the 207k-item $\times$ 768 $\rightarrow$ 300 $\rightarrow$ 64 MLP forward 1,400+ times per evaluation.

3.7 Added components (deliberately small adaptations)

Our two architectural additions (the novelty boundary is stated in §2.3) are deliberately simple, default-OFF, and zero-initialized so that each is a *bit-identical no-op at initialization* and reduces the design to a single controlled lever. Both are defined on top of an **HSTU-style pure-PyTorch implementation of the HSTU encoder, based on the published architecture** [Zhai et al. 2024]: we use the pointwise-aggregated interaction $\text{silu}(QK^\top + \text{rab})V$ with the published per-block normalization, having diagnosed and removed two spurious normalizations (a $1/\sqrt{d}$ score scaling and a $1/(i+1)$ row averaging) in our initial reimplement that collapse HSTU’s pointwise attention into weak mean-pooling. We verify the implementation equation-by-equation against the published HSTU formulation (pointwise $\text{silu}(QK^\top + \text{rab})V$ aggregation, no softmax, no $1/\sqrt{d}$ scaling); an executable end-to-end reproduction of the official kernel stack is impossible on our hardware (§5.6); the core block, however, passes an exact numerical parity test against the reference research implementation at a mirrored **aligned configuration only** (identity affine norms, zeroed extra uvqk bias, ϵ aligned, dropout off, eval mode; max abs diff 0.0; HSTU_PARITY_REPORT.md discloses the alignment — the trained configuration is a strict-superset HSTU-style variant, §6.3). **Initialization disclosure (2026-07-19)**: with the delta-kernel and gate $g=0$ initialization, both task gradients are exactly zero at the start ($y=x$, so the gate gradient vanishes and the kernel gradient is multiplied by $g=0$); learning bootstraps only because Adam’s coupled weight-decay term ($1e-5$) perturbs the delta tap off its fixed point — the optimizer regularizer is therefore part of the filter’s learning mechanism, and with weight decay disabled this parameterization is an absorbing no-op. The fixed-average and no-gate comparator arms do not share this singular start, so those comparisons alter the optimization path as well as the named design element. The later nonsingular studies provide cleaner internal contrasts but reuse outcome-visible splits, and active controls prevent uniquely learned-tap attribution; they do not retroactively decompose the non-initialization-matched historical arms.

(a) Text-Anchored Prototype Embeddings (TAPE). TAPE is a simple soft text-prototype additive capacity term inspired by semantic-ID and prototype methods (TIGER, VQ-Rec, ProtoMF); it provides a small but not headline gain (§5.1). Let $t_i \in \mathbb{R}^{d_{\text{text}}}$ be item i ’s frozen text embedding (MiniLM or BLaIR). We k-means the L2-normalized text embeddings into K centroids $\{c_1, \dots, c_K\}$ once, offline, and form a **frozen soft assignment** $A_i = \text{softmax}(\langle t_i, c \rangle / \tau) \in \Delta^K$ ($\tau = 0.05$). We add a **learnable prototype table** $P \in \mathbb{R}^{K \times d}$ (zero-initialized) to each item’s feature:

$$e_i = \text{id}_i + \Phi(t_i) + A_i P,$$

where id_i is the learned per-item embedding and Φ the learned projection of the frozen text embeddings. Items in the same semantic neighborhood share the learnable capacity $A_i P$ — the shared-structure benefit of TIGER-style semantic IDs [Rajput et al. 2023] without discrete codes or an autoregressive decoder. Because A is frozen and P is zero-init, TAPE is an exact no-op at initialization and recovers the baseline. TAPE is adjacent to TIGER (discrete RQ-VAE IDs), VQ-Rec [Hou et al. 2023] (PQ codes), and ProtoMF [Melchiorre et al. 2022] (prototype matrix factorization for explainability); it differs only in realization — a frozen text-derived soft assignment gating a learnable prototype table inside a sequential recommender — and we present it as a small adaptation of these prior ideas tested as a secondary ablation, not as a headline contribution. Default $K = 512$.

(b) Canonical causal FIR temporal filter (the proposed module). BSARec [Shin et al. 2024, Theorem 1] proves a low-pass limit for repeatedly applied softmax self-attention. That result does not cover the non-softmax HSTU-style pointwise operator used here and supplies motivation only. We test whether an explicit local left-causal residual benefits this backbone. Learned FIR separates from one equal-parameter current-position-only DCT/GELU residual, a compound contrast that does not isolate temporal access, while a competitive shared causal filter prevents attribution to per-channel taps (§5.1–§5.2). The proposed module is a single **per-channel learnable FIR residual** on the sequence embeddings $x \in \mathbb{R}^{B \times L \times d}$, in a **nonsingular, gradient-active** parameterization:

$$x \leftarrow x + \text{DWConv}_{\Delta}(\text{pad_left}(x, K-1)), \quad \Delta = 0 \text{ at init},$$

implemented as a depthwise `Conv1d(d, d, kernel_size=K, groups=d, bias=False)` whose tap residual Δ is zero at initialization, with a **left-only causal pad of $K-1$** , so the layer is an **exact identity at initialization** yet has **nonzero gradients on the first task update** — there is no gate whose zero initialization creates an absorbing fixed point. The primary E-A and canonical-breadth learned arms used the backbone weight-decay setting for FIR taps; E-A’s zero-FIR-weight-decay sensitivity was not detectably different ($A2-A1 + 0.000010 [-0.000339, +0.000360]$, $p = .95$), which is neither an equivalence result nor a pathway/mediation test (§5.2). The left-only pad makes output position t a function of input positions $\leq t$ only — so under our all-position next-item loss there is **no future-label leakage**. FMLP-Rec [Zhou et al. 2022] and BSARec use bidirectional sequence filters in their original formulations; under our all-position next-item loss, directly inserting such filters before each position’s prediction would mix future positions, so we use this left-causal realization. **Legacy zero-gated realization (historical runs; reported as package-level).** The Video_Games headline configuration (§5.1) and the original breadth campaign (§5.2) were trained with a functionally related but *singular* realization — $x \leftarrow x + g \odot (y - x)$ with a delta kernel and a scalar gate $g = 0$ at initialization — in which both task gradients vanish at the start, so learning could begin only after Adam’s weight-decay term perturbed the delta tap (§3 initialization disclosure). We call that the **legacy zero-gated FIR package** and report its contrasts as package-level (FIR-plus-initialization/optimizer), not FIR-specific. The nonsingular canonical form was then evaluated under matched per-seed initialization on three already outcome-visible category/split settings. Those internal estimates are positive, but their test exposure prevents an independent-confirmation claim. On Musical_Instruments, a later equal-parameter pointwise placebo discriminates learned FIR from that compound current-only residual without isolating temporal access, and the shared causal control prevents learned-per-channel-tap attribution (§5.2). The novelty boundary is narrow and we state it plainly. *Not new:* filtering sequential representations, causal convolution (Caser; NextItNet), the frequency-domain motivation, and the attention-oversmoothing framing (FMLP-Rec; BSARec). *New:* the minimal, gradient-active, identity-initialized, leak-free left-causal depthwise-FIR residual inserted before an HSTU-style stack under the all-position next-item objective, evaluated under full-catalog LLOO. As an FIR filter the module has a learnable per-channel frequency response — the only sense in which we use the word “spectral” — and

supplies an explicit local temporal path; we do not claim the attention stack cannot represent this behavior. The legacy-package gain is robust across kernel lengths $K \in \{4, 8, 16, 50\}$ (§5); the canonical primary configuration fixes $\mathbf{K} = 16$ (historical runs used $K = 8$ on Video_Games and $K = 16$ on Musical_Instruments, §5.1–§5.2).

A separate, subsequently frozen Software attempt also evaluated the canonical $K=16$ form and returned a positive thresholded verdict. Because earlier V2 validation-output non-visibility is not independently established, it is classified outcome-known/exploratory same-team robustness; its same-investigator, same-code-lineage, same-Amazon-family, local-custody scope also precludes independent-confirmation wording (§5.2).

Both components are leak-free by construction and add negligible parameters (TAPE: $K \times d \approx 33k$; filter: $K \times d \approx 0.5-3k$). Two further regularizers we use are *not* claimed novel: **label smoothing** [Szegedy et al. 2016] on the full-softmax CE target, and the standard time / text-similarity / relative-position attention biases (TiSASRec [Li et al. 2020]; HSTU [Zhai et al. 2024]; Shaw et al. 2018), each cited at the point of use.

4 Experiments

4.1 Datasets

The primary development and comparator studies use AR2023 5-core categories under the full-catalog LLOO protocol (§3.6). A separate prospective robustness and efficiency study uses MovieLens 1M [Harper and Konstan 2015] under a frozen global-time split. AR2023 counts are total 5-core interactions; the MovieLens row reports its filtered primary view and training-observed catalog:

Category	Role in this paper	Users	Items	Interactions (total)
Video_Games	headline reference numbers (§5.1); tail-pattern null (equivalence claim retracted §5.3)	94,762	25,612	814,586
Musical_Instruments	outcome-known FIR internal study + pre-declared per-category point-estimate comparison (§5.2)	57,439	24,587	511,836
Office_Products	V1 prereg VOID (descriptive; Appendix A.0); redesigned V3 prereg PASSED (§5.2; OFFICE_V3_RESULTS.md)	223,308	77,551	1,800,878
Beauty_and_Personal_Care	tail-pattern exploratory 3-seed estimate (§5.3); superseded cross-pipeline scan (Supplement S.1)	729,576	207,649	6,624,441
Industrial_and_Scientific	pre-declared FIR-breadth: artifact-PASS (frozen rule fired; paired premise withdrawn; post-hoc independent-arm estimate — §5.2; FIR_BREADTH_RESULTS.md)	50,985	25,848	412,947
CDs_and_Vinyl	pre-declared FIR-breadth: artifact-PASS (frozen rule fired; paired premise withdrawn; post-hoc independent-arm estimate — §5.2; FIR_BREADTH_RESULTS.md)	123,876	89,370	1,552,764
MovieLens 1M, rating ≥ 4	prospectively frozen non-Amazon FIR replication/efficiency study; negative verdict (§5.2; Harper & Konstan, 2015)	1,033	2,359 training-observed	575,281 filtered events; 135,131 train rows

Video_Games/Musical_Instruments/Office_Products match the HSTU-BLaIR comparator pipeline’s own statistics exactly on users/items (interactions within ± 1 ; §5.2). The Beauty_and_PC catalog is 8 $\times$ larger than Video_Games, with

median user history of only 5 interactions — a harder benchmark. Every pre-declared campaign that has completed is mechanically adjudicated (verdicts at the relevant result blocks: §5.2, §5.7, §5.8, and Appendix A.0).

4.2 Baselines

- Popularity (trivial floor)
- SASRec with sampled-512 softmax (our “original” baseline)
- SASRec with chunked-full-softmax (improved baseline)
- BERT4Rec [Sun et al. 2019]
- Published comparators: TIGER [Rajput et al. 2023], BLAIR [Hou et al. 2024], LIGER [Yang et al. 2024]

4.3 Experimental design (headline runs)

All headline results (§5.1–§5.4) use the HSTU-style pure-PyTorch encoder (§3.2) under the fixed AR2023 5-core full-catalog LLOO protocol (§3.6). Each configuration is trained for **40 epochs** (the §5.1 Video_Games headline; the pre-declared-file MI/Office V3/FIR-breadth campaigns train 20 epochs per their frozen configurations) at batch size 256 (d_model 64, 4 layers, 2 heads, dropout 0.5) with the memory-efficient chunked-full-softmax loss (§3.5, item chunk 32,768) and a warmup-cosine learning-rate schedule. On top of the bias stack (TAPE-512 + time bias + text-similarity bias + pos-rab) the winning configuration adds label smoothing ($\epsilon=0.2$) and the causal FIR filter ($K=8$). We run **6 seeds (20260608–20260613)** for the full-model headline and **5 seeds (20260608–20260612)** for every per-component ablation rung, reporting mean $\pm$ sample-std; the reported test number is always selected by best validation NDCG@10 (best-by-val), never by test. **Fixed-split inference scope (stated prominently, 2026-07-20)**: the training seed is the only randomized unit; every interval in this paper quantifies optimization variability on one fixed public split per category — nothing here estimates user-resampling, split-choice, category-sampling, or cross-dataset uncertainty, and all claims are scoped to these exact datasets and splits. Evaluation is full-catalog ($n_{\text{eval}} = 94,762$ on Video_Games), with all train+val items masked (§3.6).

The tail analysis (§5.3–§5.4) rests on two arm comparisons, each toggling one named configuration flag on the same split and seed numbers (bundled interventions: the arms are not initialization-paired and optimizer paths differ) and reporting by `popularity` NDCG@10 / HR@10 on leak-free train-frequency terciles (terciles frozen at full density):

- **text vs ID-only**: the full text stack against an ID-only ablation (`--no-sbert`, no prototypes, no text-similarity bias).
- **density / connectivity titration (§5.4)**: training-only sub-sampling that thins either interactions (interaction-mode) or whole users (user-mode) to match a sparser reference category’s resource levels, with the evaluation set held fixed.

A separate **20-variant cross-pipeline transfer scan on Beauty_and_PC** — supporting reproducibility material, not part of the headline spine — is reported in Appendix A.1.

The MovieLens study used six exact matched-initialization arms over eight pre-specified seed blocks: identity, channel-shared $K=16$, eight-group $K=16$, rank-at-most-8 tangent-factorized $K=16$, per-channel $K=16$, and an equal-parameter current-position-only placebo. Training constructed maps from TRAIN+VALID without opening or hashing TEST bytes. After all 96 training runs completed, each selected checkpoint received one sealed TEST evaluation. The primary estimand was NDCG@10 on rating-at-least-4 events after a global 90% time boundary; all ratings, user/item

cluster bootstraps, and resource readings were frozen sensitivities. The three-candidate noninferiority family used margin 0.000500 and Holm-adjusted one-sided tests.

4.4 Training details and hardware

Headline runs use Adam (lr 1e-3, weight decay 1e-5) with gradient clipping (norm 5.0) under the 40-epoch warmup-cosine schedule above. Hardware: a single NVIDIA RTX 5060 Ti (16 GB, Blackwell sm_120). Video_Games training takes ~10 min/seed; Beauty_and_PC ~1.5–2 hr/seed with the chunked-full-softmax loss. (The Appendix A.1 Beauty scan used a shorter 15–20-epoch budget and seeds 42/43; those details are local to that supporting study.)

5 Results

Statistical reporting conventions. Unless stated otherwise: the sample unit is the training seed; means are reported $\pm$ sample standard deviation over seeds; confidence intervals are two-sided 95% Student-t intervals with $df = n-1$; comparisons between our own arms use same-numbered seeds and are analyzed as independent arms (Welch/Satterthwaite) unless exact matched backbone initialization and a paired analysis are explicitly named, as in canonical breadth, active controls, pointwise placebo, and prospective Software V3; comparisons against published numbers are point-estimate comparisons; pooled tail hit-counts are descriptive counts only; p-values are uncorrected unless a correction is named. Cells labeled *confirmatory* in the artifact manifest are pre-declared prospective campaigns or frozen-protocol locks: selection timing and protocol integrity, not seed count or external independence, are the criterion. Multi-seed post-hoc results remain exploratory.

5.1 Video_Games — multi-seed reference numbers (NOT SOTA)

Our headline result (Table 1) is built on the **HSTU-style pure-PyTorch encoder** with the two added components and stacks each lever in a controlled 5-seed ablation; Table 1a retains the traceable within-paper SASRec-family floor; Table 1b combines reported HSTU-BLaIR evidence, our SM120 compatibility port, an official-code WEARec equal-evaluation feasibility run, and the frozen AlphaFuse-style whole-package comparison.

Table 1: Headline component ablation — NDCG@10 on AR2023 Video_Games 5-core LLOO (full-catalog eval, $n_{eval} = 94,762$; 5 seeds = 20260608...20260612 unless noted; the full-model row is 6-seed 20260608...20260613; one flag added per row).

Configuration	NDCG@10	seeds	Δ
HSTU-style encoder, plain	0.0588	1	—
+ TAPE-512 (sub-additive text component)	0.0597	1	+0.0009
+ full bias stack (TAPE + time + text-sim + pos-rab)	0.0637 ± 0.0003	5	+0.0049 vs plain (+8.3%)
+ label smoothing $\epsilon=0.2$ [Szegedy et al. 2016]	0.0649 ± 0.0003	5	+0.0012 (bands non-overlapping)
+ causal FIR filter K=8 (ours) → full model	0.0673 ± 0.0003	6	+0.0024 (bands non-overlapping)
(single-arm) FIR package arm (filter component; attribution open), no label smoothing	0.0652 ± 0.0003	5	+0.0015 vs the 0.0637 stack
(single-arm) ID-only (no SBERT / no text-sim / no prototypes)	$\approx 0.0656 \pm 0.0002$	5	text adds only +0.0018 (+2.7%) overall

The full model is **+17.5%** over the published SASRec point estimate (0.0573, Table 1b) on the same reported Video_Games dataset geometry, but published training/evaluation equivalence is not established — and **this difference is architectural, not text-driven**: the ID-only ablation already reaches ≈ 0.0656 (itself +14% over published SASRec), and the entire frozen-text stack (SBERT features + text-sim bias + prototypes) adds only **+0.00178 $\pm$ 0.00021 (5-seed, +2.7%)** overall. The two supported regularizers stack with a combined lift whose excess over the component sum is 33% — additivity and orthogonality were NOT tested (loss target vs. embedding spectrum): label smoothing alone +0.0012, causal filter alone +0.0015, combined +0.0036 over the bias stack. Per-component single-flag attribution on the HSTU-style implementation: the **time bias is the largest classical component (+0.0027)**, label smoothing +0.0013, causal filter +0.0015, while **text-similarity bias shows no observed benefit (± 0.0001 , removable candidate)** and **TAPE is sub-additive (+0.0009)**. *(The time-bias, text-sim, and TAPE single-flag figures here are the single-seed seed-20260608 DECOMP values, honestly flagged $n=1$ (reported inline here, not in a table). A 4-seed multi-seed cross-check — DECOMP5, seeds 20260609–12, HSTU-style base 0.0594 — confirms the ordering: time bias +0.0030 (largest classical component, $\geq$ the single-seed +0.0027), pos-rab +0.0012, TAPE +0.0004, text-sim -0.00005 (no benefit observed at tested power). The qualitative attribution is unchanged; the only number that materially moves is TAPE, whose multi-seed single-flag lift (+0.0004) is smaller still than the single-seed +0.0009 — further support for its demotion from a headline component.)* The causal filter’s gain is robust across kernel lengths $K \in \{4, 8, 16, 50\}$ (k16 marginally best & tightest, 5-seed 0.0676 ± 0.0002 ; low kernel sensitivity, §5.4). Every capacity-adding probe we tried instead (continuous time-decay kernel, expert heads, text-distillation, dual-text, EMA, CL4SRec, James–Stein shrinkage [Efron and Morris 1973; James and Stein 1961], niche-competition loss, heat-kernel label smoothing, spectral-shrink, cue-fusion, forced ID→text routing) was neutral or harmful — several with a learned scalar the model itself drove to zero (the negative-result map, tabulated in Table S1, §5.5).

Table 1a: Traceable within-paper SASRec-family floor (full-catalog eval).

Method	NDCG@10	HR@10	Notes
popularity	0.0125	0.0248	trivial floor (traceable: results_5core_Video_Games.json)

The v1-era SASRec/SBERT/BLaIR baseline rows and the observations derived from them have been REMOVED per the artifact-graph gate: their per-seed artifacts were not retained and the values are untraceable. Exact user/item counts, ± 1 interaction agreement, frozen split hashes, and per-category floor runs establish dataset identity and local split traceability only; they do not establish training, implementation, or evaluation parity with a published system.

Table 1b: Reported, locally regenerated, and current-comparator evidence [Hu et al. 2025; Liu 2025; Xu et al. 2026]:

The HSTU-BLaIR paper reports on AR2023 Video_Games 5-core LLOO with identical dataset stats to ours (25,612 items / 94,762 users / 814,585 interactions). They train for **100 epochs** following Zhai et al.’s HSTU protocol. Their single-seed numbers:

Method	NDCG@10	HR@10	Comparison to ours
SASRec [Liu 2025, single seed, 100 epochs]	0.0573	0.1028	−15% below our full model 0.0673 ± 0.0003
HSTU [Zhai et al. 2024]	0.0741	0.1315	+10% above our full model
HSTU-OpenAI (TE3L)	0.0742	0.1328	+10% above our full model
HSTU-BLaIR [Liu 2025]	0.0760	0.1353	+13% above our full model; stronger reported reference
HSTU-BLaIR local SM120 compatibility port (this audit)	0.07382 final full-eval (exported artifact; the higher best-epoch reading survives only in an unretained WSL log and is excluded from the artifact graph)	0.13234 final	stronger than ours; not a faithful pinned-environment reproduction
AlphaFuse-style MiniLM representation package (8 fresh seeds; this audit)	0.04827 [0.04813, 0.04842]	0.08976 [0.08935, 0.09016]	EEV3-REPORTABLE-OUTCOME-KNOWN; +0.00925 [0.00833, 0.01017] vs repository SASRec-ID, but −0.01906 [−0.01935, −0.01878] vs our reference
AlphaFuse-repository SASRec-ID backbone (8 fresh seeds; this audit)	0.03902 [0.03811, 0.03994]	0.07363 [0.07235, 0.07491]	frozen V3 ID-only comparator; not paired with the package arm
WEARec official model/training code (8 assessment seeds; this audit)	0.05918 [0.05867, 0.05969]	—	WEAREC-BELOW-EXISTING-REFERENCE; −0.00815 [−0.00869, −0.00762] vs our six-seed 0.06734 reference under the shared evaluator

Our full model (0.0673) exceeds their published SASRec (0.0573) but does not reach their HSTU (0.0741) or HSTU-BLaIR (0.0760) on Video_Games. Our local compatibility-port HSTU-BLaIR run reached final full-eval NDCG@10 0.07382 (the best-epoch reading exists only in an unretained WSL log and is excluded from the artifact graph); the run is stronger than SASRec-SBERT but carries SM120/fbgemm validity caveats.

The separately frozen WEARec campaign used the official AAAI 2026 repository at commit 2087335339b1 (the full hash is artifact-bound). Two official-domain presets were selected by one tuning seed using validation NDCG@10 only; the selected official_beauty preset then trained on eight fresh assessment seeds with TEST scoring suppressed until all checkpoints existed. The committed adjudicator’s first authorized endpoint read returned **WEAREC-BELOW-EXISTING-REFERENCE**: WEARec mean NDCG@10 **0.059184**, 95% CI [**0.058674**, **0.059693**], versus the existing six-seed full-model reference **0.067337** [**0.067063**, **0.067611**]. The descriptive outcome-known unpaired Welch contrast is **−0.008154** [**−0.008689**, **−0.007618**], $p = 1.16 \times 10^{-11}$. This is narrow official-model/equal-evaluation evidence by the same investigators on an outcome-known split—not independent confirmation, SOTA, a paired experiment, or

equality of architecture, loss, schedule, or tuning budgets. The graph replays the released NDCG vectors and Welch arithmetic; private endpoint extraction and HR/MRR raw-vector arithmetic are not publicly replayed.

The clean E-E V3 campaign froze the AlphaFuse upstream at commit b501a0540b60 (the full hash is artifact-bound), the official repository AlphaFuse and SASRec classes, one shared training configuration, and eight fresh seeds per arm. MiniLM-384 title vectors replace AlphaFuse’s published OpenAI text vectors, so this is an AlphaFuse-style port rather than a published-table reproduction. All 16 checkpoints were selected using complete-history-masked VALID NDCG@10 before any TEST read; 16 sealed one-shot TEST evaluations then preceded the unchanged committed adjudicator. Its exact verdict, **EEV3-REPORTABLE-OUTCOME-KNOWN**, is countable as current-comparator evidence: AlphaFuse-style NDCG@10 **0.048273 [0.048129, 0.048416]**, repository SASRec-ID **0.039024 [0.038106, 0.039941]**, and descriptive independent-arm Welch difference **+0.009249 [+0.008329, +0.010169]**, $p = 3.48 \times 10^{-8}$. Fixed-split user and target-item-cluster bootstrap sensitivities are also positive, but are not optimizer- or population-level inference. The package remains **-0.019065 [-0.019347, -0.018783]** below the existing 0.067337 full-model reference. Because text availability, initialization, trainable capacity, parameter allocation, and architectures differ, this is a whole representation-package contrast—not null-space-fusion isolation, equal-tuning evidence, a paired experiment, independent confirmation, or SOTA. The graph recomputes released aggregate vectors and Welch arithmetic and validates all endpoint/sidecar hashes; it does not replay private endpoint extraction or record-level bootstraps.

Relative to HSTU-BLaIR, this paper contributes multi-seed internal ablations, a released preprocessing path, and implementation diagnostics—not a leaderboard result. The reported 0.0760 remains the stronger Video_Games reference. Core-block equality at one aligned configuration (§3.2) and environment-caveated local regenerations (§5.6) do not constitute a pinned end-to-end reproduction.

Older published baselines on different protocols (NOT directly comparable; for context only):

Method	NDCG@10	Dataset	Protocol
SASRec (in TIGER paper)	0.0318	Amazon Beauty 2014	their 5-core (different category, different dataset year)
TIGER [Rajput et al. 2023]	0.0384	Amazon Beauty 2014	their 5-core
SASRec (in LIGER paper)	0.02179±0.00023	Amazon Beauty 2014	their 5-core
LIGER (K=20, Yang et al. [2024])	0.04020±0.00044	Amazon Beauty 2014	their 5-core
LIGER (K=N, full retrieval)	0.04738±0.00151	Amazon Beauty 2014	their 5-core
Best LLM-encoder in BLaIR (Hou et al. [2024]) Table 8	0.0138	AR2023 Video_Games	by-timestamp 8:1:1, no k-core , UniSRec downstream

Comparability caveat: TIGER and LIGER evaluate on the older Amazon Reviews 2014 dataset, not AR2023. The BLaIR paper does evaluate on AR2023 Video_Games but uses a by-timestamp split (no k-core filter) and a different downstream architecture (UniSRec), producing numbers ~3-4× lower than our 5-core LLOO numbers. The 5-core filter guarantees ≥ 5 interactions per user and item in the full filtered sequence (which substantially eases the benchmark relative to the by-timestamp variant); note that after the leave-last-out split a small set of items retains zero *training* occurrences (§5.3 cohort disclosure). We therefore do **not** claim SOTA over TIGER/LIGER/BLaIR. HSTU-BLaIR is the relevant stronger AR2023 Video_Games 5-core reference and blocks a SASRec-SBERT SOTA claim.

Concurrent protocol landscape. SID-MLP [Guo et al. 2026, arXiv:2605.12617], Latte [Hou et al. 2026a, arXiv:2605.06331], and GrIT [Shyam et al. 2026, arXiv:2602.19728] report same-statistics AR2023 LLOO results; ReSID [Liang et al. 2026, arXiv:2602.02338] and ChronoSID [Huang et al. 2026, arXiv:2607.03918] use a different filtered universe; *Augment or Not?* [Huang et al. 2025, arXiv:2505.23053], DiffuReason [Jiang et al. 2026, arXiv:2602.09744], and SILLM4Rec [Wu et al. 2025, MMAAsia 2025] use non-interchangeable filtering or sampled-ranking setups; UniSGR [Sun et al. 2026, arXiv:2607.04068] uses private logs; and DIGER [Fu et al. 2026, arXiv:2601.19711] and ACERec [Xia et al. 2026, arXiv:2602.13573] address related semantic-ID problems outside this exact public protocol. We cite these works for coverage and make **no comparison against concurrent preprints**. HSTU-BLaIR remains the published external anchor for the counted point-estimate comparisons (§5.2).

Within our own full-catalog runs, the 11.6M-parameter model trains in about 10 minutes on one consumer GPU and reaches $\approx 5.4\times$ the popularity floor (0.0125). These are efficiency and sanity-check observations, not priority or SOTA claims.

5.2 Canonical causal FIR internal contrasts and untuned category reuse (legacy-package attribution caveat §3)

The strongest second-category package result appears on Musical_Instruments, an already outcome-visible AR2023 category; it is internal robustness evidence, not transfer confirmation. The full V2 stack reaches **NDCG@10 = 0.0413 $\pm$ 0.0005 (5-seed)**, +7.9% over a matched HSTU+TAPE base (0.0383 $\pm$ 0.0004) and +57% over a plain ID-only SAS-Rec (0.0264). A two-arm comparison (not a controlled isolation — §3 initialization caveat) (best-by-val, vs the MI SBERT+TAPE base 0.0383, four 20-epoch seeds 20260609–12) separates the two supported regularizers at the arm level (bundled arms; no component isolation):

Table 1c: Musical_Instruments arm-by-arm comparison (NDCG@10, best-by-val, vs MI SBERT+TAPE base; shares are of the 5-seed V2 combined lift +0.0032).

Configuration	NDCG@10	Δ vs base	share of combined lift
MI SBERT+TAPE base (four 20-epoch seeds)	0.0383	—	—
+ label smoothing only (5-seed)	0.0391 $\pm$ 0.0001	+0.0008	26%
+ FIR package arm (filter component; attribution open) (k16, 5-seed)	0.0408	+0.0025	77%
+ both = V2 (k16, 5-seed)	0.0415	+0.0032	100%

(Recomputed best-by-val from the released per-seed artifacts. Against the headline 5-seed V2 stack the package arm’s share is 77% (k16) / 82% (k8). LS-only is the 5-seed family results_MI_lsonly_seed{08–12} (0.03914 $\pm$ 0.00013); base is the four e20 seeds 20260609–12.)

The package arm contributes $\approx 3\times$ what label smoothing does on the second category, and $\approx 80\%$ of the combined lift in that same-seed decomposition — a package-level share, no component isolation — (77% vs the k16 stack, 82% vs the k8 headline stack) — i.e. the historical package’s gain appears on both development categories while label smoothing’s does not (no formal component-by-category interaction was tested). Its kernel-length insensitivity also appears on MI (MI k8 0.0413 $\pm$ 0.0005 $\approx$ k16 0.0415 $\pm$ 0.0002). Component attribution for these historical runs remains open; the matched-initialization studies below provide cleaner internal contrasts, subject to their outcome-visibility limits.

Pre-declared matched-arm FIR study (PREREG_FIR_V3.md / E-A; mechanically adjudicated verdict W-POS; outcome-visible split). A nonsingular reparameterization, $y = x + \text{conv}_\Delta(x)$ with $\Delta = 0$, made the treatment identity-initialized and gradient-active; three arms by eight seed blocks shared one verified initialization within each block. The frozen registered analysis is independent-arm Welch/Satterthwaite: learned FIR minus identity is **+0.002265, ordinary 95% Welch CI [+0.001928, +0.002602]** ($df = 13.939$) on Musical_Instruments. The paired-by-seed difference is retained only as descriptive. This is a cleaner within-configuration contrast than the historical package arms, but the category and split had already informed development and the execution record contains the provenance deviations documented in the audit; it is therefore exploratory internal evidence, not fresh confirmation. The zero-FIR-weight-decay sensitivity is A2-A1 **+0.000010 [-0.000339, +0.000360]**, $p = .95$: no difference was detected, but neither equivalence nor exclusion of a weight-decay pathway is established. Artifacts: results_MI_FIRV3_{arm}_seed{seed}.json (24 runs) and fir_v3_adjudication.json.

Pre-declared canonical reparameterization breadth (PREREG_FIR_CANONICAL_BREADTH.md; verdict CANON-BREADTH-POS; outcome-known/test-exposed). With one frozen K=16, 20-epoch configuration, zero category tuning, and exact per-seed matched initialization, learned minus identity is **+0.002110, ordinary paired 95% CI [+0.001820, +0.002399]** on Industrial_and_Scientific (paired $t = 17.23$; Holm-adjusted $p = 5.44 \times 10^{-7}$; 8/8 positive) and **+0.006150 [+0.005849, +0.006450]** on CDs_and_Vinyl ($t = 48.39$; Holm-adjusted $p = 8.42 \times 10^{-10}$; 8/8 positive). Supportive registered endpoints point in the same direction: HR@10/MRR differences are +0.003474/+0.001819 on Industrial_and_Scientific and +0.010728/+0.005033 on CDs_and_Vinyl; the broader registered cutoff family was not emitted by the trainer and is not claimed. The categories were selected after favorable legacy-package outcomes and TEST was evaluated each epoch, so these estimates are **descriptive robustness of the canonical parameterization, not independent confirmation or population transfer**. The intervals are ordinary paired-t intervals; Holm adjusts decisions/p-values, not confidence limits. All 32 source JSONs and the adjudication record are now hash-bound in the public artifact inventory.

Pre-declared six-arm active-control study (PREREG_FIR_CONTROLS.md; eight matched seed blocks; sealed one-shot final TEST; frozen verdict CTRL-ACTIVE-CONTROL-SUPPORTED). The frozen analysis has two separate Holm families; all intervals are ordinary paired 95% intervals. In family A, learned is **+0.002116 [+0.001910, +0.002322]**, $p_{\text{Holm}} = 2.04 \times 10^{-7}$; fixed MA **+0.000712 [+0.000509, +0.000915]**, $p_{\text{Holm}} = 1.30 \times 10^{-4}$; fixed HP **+0.000708 [+0.000510, +0.000907]**, $p_{\text{Holm}} = 1.30 \times 10^{-4}$; channel-shared **+0.002197 [+0.002008, +0.002386]**, $p_{\text{Holm}} = 1.08 \times 10^{-7}$; and parameter-matched nonlinear causal **+0.001912 [+0.001657, +0.002166]**, $p_{\text{Holm}} = 1.33 \times 10^{-6}$. In family B, learned exceeds fixed MA by **+0.001404 [+0.001085, +0.001723]**, $p_{\text{Holm}} = 6.37 \times 10^{-5}$ and fixed HP by **+0.001407 [+0.001089, +0.001726]**, $p_{\text{Holm}} = 6.37 \times 10^{-5}$, but the fixed arms are algebraically redundant. A learned-specific advantage is not established over shared (**-0.000081 [-0.000337, +0.000175]**, $p_{\text{Holm}} = .477$) or nonlinear (**+0.000204 [-0.000038, +0.000446]**, $p_{\text{Holm}} = .174$); retained tests are not equivalence. This study alone lacked a parameter-matched non-temporal placebo. It was designed after the learned-identity outcome was known, ran from a dirty evolving tree, and its sidecars were not independently custodied because of a filename-suffix ignore error; we classify it as outcome-known exploratory mechanism evidence despite its frozen analysis and sealed evaluator. Human pre-adjudication visibility remains an author-verification item.

Pre-declared parameter-matched non-temporal placebo (PREREG_FIR_POINTWISE_V1.md; eight matched seed blocks; sealed one-shot final TEST; frozen verdict POINTWISE-FIR-DISCRIMINATED). The current-position-only placebo uses a deterministic width-16 orthonormal DCT projection, GELU, and zero-initialized learned 16-to-64 map. It is identity-initialized, gradient-active, and has exactly 1,024 trainable parameters, matching the K=16 depthwise

FIR, but cannot access earlier positions. In one frozen three-test Holm family, learned FIR minus identity is **+0.001872** [**+0.001737, +0.002007**], $p_{\text{Holm}} = 1.87 \times 10^{-8}$; pointwise minus identity is **-0.000069** [**-0.000200, +0.000061**], $p_{\text{Holm}} = .249$; and learned FIR minus pointwise is **+0.001941** [**+0.001788, +0.002095**], $p_{\text{Holm}} = 2.40 \times 10^{-8}$. All intervals are ordinary paired 95% intervals; the retained pointwise-identity test is not equivalence. This discriminates the learned temporal FIR from the tested equal-parameter non-temporal residual under matched initialization. It does not establish per-channel-tap necessity because the shared causal filter remains competitive, and it is outcome-known Musical_Instruments mechanism evidence, not independent confirmation or cross-domain replication.

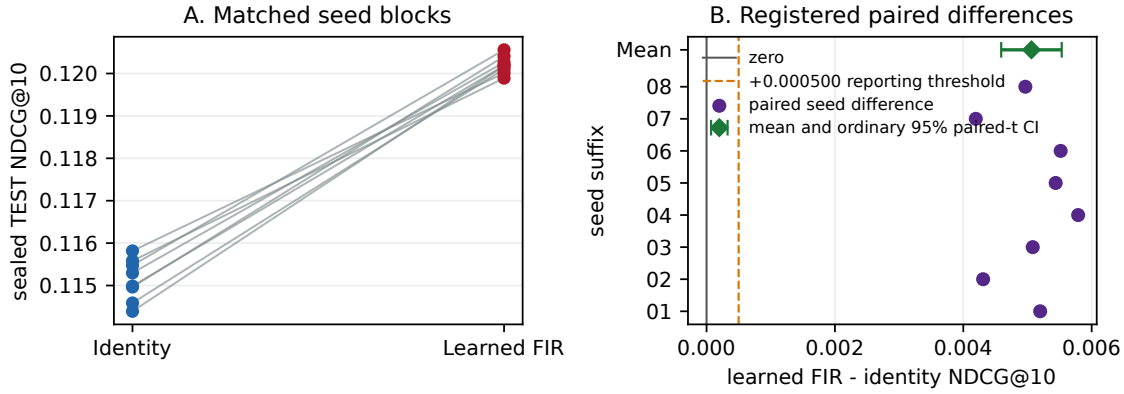
Frozen Software robustness attempt (PREREG_FIR_PROSPECTIVE_SW_V3.md; eight matched seed blocks; sealed one-shot final TEST; verdict SW-V3-PRACTICAL-POS). The learned and identity arms used exact per-seed matched backbone initialization. Training loaded TEST rows only for the explicitly pre-declared transductive item catalog and emitted no TEST scores; checkpoints were selected solely by validation NDCG@10. After all 16 checkpoints existed, the committed driver created exclusive-created, hash-linked local READY/evaluation seals, evaluated each selected checkpoint once on TEST, and invoked the protocol-designated adjudicator. Mean TEST NDCG@10 was **0.120200 learned versus 0.115138 identity**; learned minus identity was **+0.005062, ordinary paired 95% CI** [**+0.004591, +0.005533**], paired $t(7) = 25.39$, two-sided $p = 3.75 \times 10^{-8}$, with 8/8 differences positive. The CI lower bound cleared the frozen +0.000500 reporting threshold, yielding the exact mechanical verdict SW-V3-PRACTICAL-POS. Tracked evidence cannot establish whether earlier V2 validation output was observed before V3 was frozen, and local logs cannot prove first human/tool access; absent an author custody statement, we therefore classify V3 as **outcome-known exploratory same-team robustness**, not confirmation. It is also same-investigator, same-code-lineage, and same-Amazon-family work under local same-user custody with no external escrow. The frozen tag additionally binds a CRLF raw digest for one MI lineage-reference JSON while a normal tagged checkout produces LF bytes; direct clean-tag execution therefore fails that raw input assertion until the historical CRLF representation is restored. The reference was not read for runtime configuration, so the defect does not change endpoint arithmetic, but it narrows replayability (FIR_PROSPECTIVE_SW_V3_REPLAY_ERRATUM.md).

The paired-difference sample SD was **0.000563779**. In registered seed order, the eight learned-minus-identity differences were **+0.005197548, +0.004308974, +0.005082181, +0.005788969, +0.005439290, +0.005516648, +0.004194543, and +0.004966712**. A post-hoc two-sided exact sign test for 8/8 positive differences gives $p = .0078125$. This sign test is a small- n robustness sensitivity, not the frozen decision rule and not population-level uncertainty across users, cutoffs, categories, or domains.

Prospectively frozen non-Amazon replication and efficiency study (PREREG_FIR EFFICIENCY_ML1M_V1.md; MovieLens 1M; verdict ML1M-NO-FIR-REPLICATION). The protocol and complete execution chain were committed and pushed before acquiring MovieLens. The rating-at-least-4 primary view used a global 90% time boundary, a training-observed catalog, 1,033 users, 2,359 items, and 135,131 training rows. All 96 checkpoints were trained before any TEST file was opened or hashed; 96 sealed one-shot evaluations then preceded the protocol-designated adjudicator's first endpoint read. This is prospectively frozen same-investigator non-Amazon evidence, not independent confirmation or population-wide generalization.

The exact primary verdict is negative. Learned FIR did not replicate against identity (**+0.000000, ordinary paired 95% CI** [**-0.000074, +0.000075**], $p_{\text{Holm}} = .995$) or the equal-parameter pointwise arm (**+0.000035** [**-0.000057, +0.000127**], $p_{\text{Holm}} = .796$). Mean NDCG@10 was 0.052151 for both learned and identity and 0.052116 for pointwise. The all-ratings sensitivity agreed: learned minus identity +0.000012 [-0.000107, +0.000131] and learned minus pointwise

Software V3: outcome-known same-team robustness (8 matched seeds)



All points are registered sealed endpoints. Evidence class: exploratory; not independent confirmation.

Fig. 1. Software V3 registered paired-seed results. (A) Every matched seed block has higher sealed TEST NDCG@10 under learned FIR than identity. (B) All eight paired differences, the frozen +0.000500 reporting threshold, and the ordinary paired-t mean and 95% CI. The figure visualizes outcome-known exploratory same-team robustness, not independent confirmation; exact plotted values are in figures/fig_software_v3_pairs_data.csv.

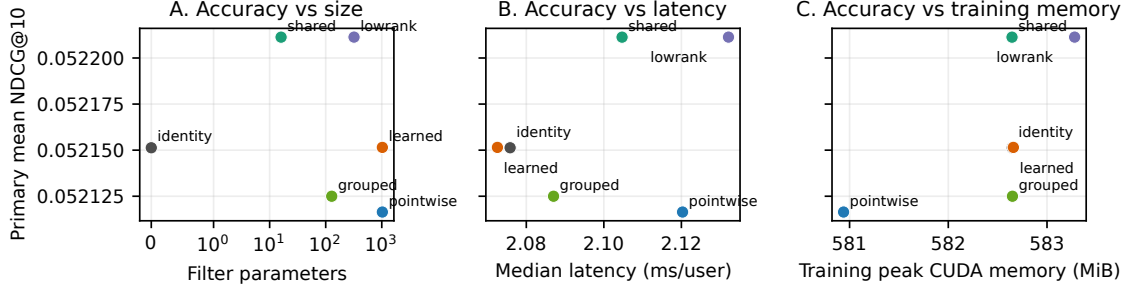
Table 2. MovieLens 1M R4 primary FIR replication and conditional parsimony. NI-PASS uses the frozen 0.000500 noninferiority margin versus learned FIR. Because the learned-FIR replication gate failed, NI-PASS is conditional numerical compression, not evidence of FIR utility.

arm	filter parameters	mean NDCG@10	arm – learned	simultaneous lower bound	frozen interpretation
identity	0	0.052151	–0.000000	—	learned replication failed
shared K=16	16	0.052211	+0.000060	–0.000068	NI-PASS
grouped K=16	128	0.052125	–0.000027	–0.000128	NI-PASS
low-rank K=16	320	0.052211	+0.000060	–0.000033	NI-PASS
learned per-channel K=16	1,024	0.052151	0	—	effect gate failed; no replicated FIR benefit
pointwise placebo	1,024	0.052116	–0.000035	—	learned–pointwise failed

+0.000043 [–0.000082, +0.000168]. All six pre-declared user/item cluster percentile intervals for the parsimonious-arm contrasts included zero.

Shared, grouped, and low-rank passed the frozen noninferiority family ($p_{\text{Holm}} = 5.20 \times 10^{-6}$, 5.20×10^{-6} , and 1.43×10^{-6}). These are conditional compression findings, not evidence that a smaller FIR is useful, because the prerequisite learned-FIR replication contrasts failed. Median latency per user was 2.076/2.105/2.087/2.132/2.073/2.120 ms in table order; primary-view training peak memory ranged from 580.94 to 583.28 MiB on one RTX 5060 Ti. The resource plane is descriptive hardware evidence, not a general efficiency ranking. Under the ML-1M README, record-level rows, checkpoints, endpoints, and per-user sidecars remain private; the public graph verifies frozen code hashes and recomputes all reported aggregate-vector arithmetic but cannot independently replay private endpoint extraction.

MovieLens 1M R4: aggregate FIR accuracy-resource plane (8 matched seeds)



Learned FIR did not replicate versus identity or pointwise; resource readings are descriptive on one GPU.

Fig. 2. MovieLens 1M R4 aggregate FIR accuracy-resource plane. Learned FIR failed to replicate versus identity and pointwise. Resource readings are descriptive on one GPU; exact aggregate values are in figures/fig_fir_efficiency_m1m_v1_data.csv.

Pre-declared breadth (two further categories). Under PREREG_FIR_BREADTH.md (committed before any run; fresh never-inspected seeds 20260713–17; mechanical adjudication in FIR_BREADTH_RESULTS.md), the filter was transplanted with **zero per-category tuning** and tested as a same-seed 5-seed filter-vs-no-filter contrast — an internal contrast with no comparator involved — on two categories never used in its development. **Both categories fired the frozen decision rule:** Industrial_and_Scientific same-seed $\Delta = +0.0024 \pm 0.0005$ (pre-declared paired-analysis 95% CI [+0.0018, +0.0030]; 5/5 same-seed differences positive) and CDs_and_Vinyl same-seed $\Delta = +0.0057 \pm 0.0006$ (pre-declared paired-analysis 95% CI [+0.0049, +0.0064]; 5/5). **Analysis-premise correction (2026-07-19):** the frozen rule *as written* is a paired Student-t, but its pairing premise is now known to be false — same-numbered seeds are not initialization-paired (§5.3 randomization disclosure) — so the pre-declared intervals are reported as executed-as-frozen outputs whose *paired interpretation is withdrawn*; the **primary supported statement** is the post-hoc independent-arm Welch analysis (we no longer call it conservative: with correlated arms neither interval has a general conservative ordering, and here the Welch intervals are in fact narrower than the paired ones), under which both effects remain positive with 95% CIs excluding zero (Industrial_and_Scientific [+0.0019, +0.0029], CDs_and_Vinyl [+0.0050, +0.0063]; graphed as firb.*.welch, evidence-class exploratory since the Welch analysis is post-hoc). Per the frozen claim wording as narrowed by this correction, each result is exactly: *the causal-FIR-filter arm's 5-seed improvement over the no-filter arm on that category is positive with a 95% CI excluding zero* — an internal same-seed contrast of the FIR-plus-initialization/optimizer package; nothing broader, no SOTA language, no comparator statement. The filter package is thus multi-seed-supported on **four categories**, two of them under this pre-declared breadth campaign — and now **estimated under the valid independent-arm TFFV2 pre-declaration (outcome-visible; not confirmatory — §5.3(vii))** (below), which supersedes this campaign's withdrawn paired interpretation as the primary FIR evidence.

TFFV2 independent-arm replication (pre-declared, outcome-visible; adjudicated 2026-07-20). The PREREG_TAIL_FIR_V2.md campaign (Git-committed frozen rules; outcome-visible — OTS chronology in §5.3, disclosure (vii)) reran both breadth categories as fully independent 8-vs-8 arms (fresh never-inspected seeds; no pairing premise;

per-run sidecars released): Industrial_and_Scientific overall $\Delta = +0.002131$ ($t = 16.99$, 95% CI $[+0.001862, +0.002400]$) and CDs_and_Vinyl overall $\Delta = +0.005770$ ($t = 26.12$, 95% CI $[+0.005275, +0.006266]$) — **both PASS under Holm** (one family with the tail endpoint E1, §5.3). This replaces the withdrawn paired interpretation with a valid pre-declared independent-arm design; the treatment remains the FIR-plus-initialization/optimizer package. Descriptively, the filter’s positive-frequency-tail improvements are $+0.000775$ (IS, $p = 2 \times 10^{-4}$) and $+0.002329$ (CDs, $p = 3 \times 10^{-12}$).

Comparator ablations (audit-requested). Each arm targets one named design element of the filter (with the disclosed caveat that the singular headline initialization means these arms also alter the optimization path, §3) on the same 5-seed MI stack: freezing the kernel at a causal moving average (learnable gate retained) recovers only $+0.00133$ of the filter’s $+0.00225$ gain over the no-filter baseline (0.04046 ± 0.00022 vs 0.04138 ± 0.00053 ; non-overlapping bands) — an observation confounded by the shared singular start (§3): no learned-shape attribution is claimed. Freezing the gate at 1 (delta-initialized kernel) matches the full filter (0.04127 ± 0.00046), so the zero-init gate comparison is confounded by the singular-initialization bootstrap (§3 disclosure) and is not a clean single-factor attribution.

Pre-declared confirmation vs the published per-category reference. The seeds above (20260608–17) are *exploratory*: kernel choice, epoch budget, and the decision to compare against the published Musical_Instruments HSTU-BLaIR number (0.0406; Liu 2025, Table 2) were all made while inspecting them. To make the comparison selection-free, we froze a version-controlled pre-declared protocol (SOTA_CONFIRM_PREREG_V2.md, committed to version control before any confirmation run: exact config and commands, data SHA256s, decision rule, claim wording) and ran **five pre-specified never-inspected seeds (20260618–22) under each of the two kernels — ten arm-runs** with per-run provenance manifests (command, git commit + clean-tree flag, environment, data hashes) and per-user record sidecars. Result: **K=16 fresh 5-seed 0.04152 ± 0.00045 (95% CI lower bound 0.04096) and K=8 0.04120 ± 0.00030 (CI-LB 0.04083) — both CI lower bounds above the published 0.0406, 10/10 seeds above.** The supported claim is exactly this: *fresh multi-seed means and seed-level confidence intervals exceed the published HSTU-BLaIR point estimate on this category under our reproduced protocol (parity evidence: users/items match the paper exactly; interactions differ by one — see SOTA_CONFIRM_PREREG_V2_ERRATA.md E1; our plain SASRec floor is below their published SASRec, ruling out an easier split).* The comparator is single-seed; our post-hoc local regeneration of it (§5.6 — the reference implementation’s research path under data-movement shims, unpinned environment) regenerates the published value at its best full-eval epoch (0.0406; final epoch 0.0391) but is itself a single environment-caveated run, so no paired or distributional superiority is claimed, and this remains a per-category result — Video_Games is explicitly not claimed (§5.1). The first execution of this confirmation was voided by its own provenance tripwire (a concurrent documentation edit dirtied the tracked tree mid-campaign) and re-executed in full under a clean tree; both executions agree per-seed to ± 0.0003 (SOTA_CONFIRM_V2_RESULTS.md). One protocol deviation is disclosed rather than claimed away: the gated runs executed at a documentation-only descendant commit of the pre-declaration’s introducing commit (the pre-declared commit-equality rule was not literally satisfied); code identity across the two commits is demonstrated by empty protocol-file diffs and, in the from-scratch rebuild, by code hashes embedded in each run manifest (SOTA_CONFIRM_PREREG_V2_ERRATA.md, E3).

§5.2 (regenerated): pre-declared dual-kernel V2 confirmation (SOTACONF_V2, fresh seeds 20260618–22, EXEC2 gated artifacts, $n_{eval}=57,439$).

kernel	fresh 5-seed NDCG@10	95% CI lower bound	vs published 0.0406
K=16	0.04152 $\pm$ 0.00045	0.04096	ABOVE
K=8	0.04120 $\pm$ 0.00030	0.04083	ABOVE

Fresh seeds above 0.0406: 10/10. EXEC1 $\leftrightarrow$ EXEC2 max per-seed $|\Delta| = 0.000057$ (< 0.0003). Clean rebuild (rebuild_v2/): K=16 CI-LB 0.04092, K=8 CI-LB 0.04084 — dual gate PASS.

Office_Products V1 (the original second-category attempt) — descriptive only; superseded by the counted V3 campaign below. Office_Products numerically exceeded the published HSTU-BLaIR point estimate under the frozen MI configuration, but the pre-declared SASRec floor check failed because our local SASRec floor was substantially above the published SASRec reference. We therefore treat the V1 Office campaign as descriptive evidence, not as a passed confirmatory category (the separately pre-declared V3 campaign later passed and is counted, §5.2). The floor anomaly has since been resolved mechanistically — the reference implementation’s own SASRec, run locally on its own pipeline (§5.6), lands +13.9% above its published row, while a completed run of their Office HSTU-BLaIR configuration regenerates its published row (+1.6%; §5.6) — the conservatism is a property of the published Office SASRec row specifically. The VOID is retained on procedural grounds (the pre-declared floor check failed as written); full detail, decomposition, and disclosures in Appendix A.0.

Office_Products V3 (redesigned pre-declaration) — PASSED. Under PREREG_OFFICE_V3.md (committed before any run; fresh never-inspected seeds 20260728–32; gate: each arm’s fresh 5-seed 95% CI lower bound must exceed the **environment-matched local regeneration of the comparator** — best full-eval 0.0279, itself above the published 0.0271 — with $\geq 4/5$ seeds individually above), **both kernel arms passed**: K=16 fresh 5-seed **0.03047 $\pm$ 0.00011 (CI-LB 0.03033)** and K=8 **0.03029 $\pm$ 0.00005 (CI-LB 0.03024)**, 10/10 seeds above both references; dataset identity, reference-artifact hashes, and per-run provenance verified mechanically (OFFICE_V3_RESULTS.md). Per the frozen claim wording: *a per-category point-estimate comparison against the published 0.0271 and its environment-matched single-run local regeneration 0.0279; no paired or distributional superiority is claimed; this is not SOTA on Office_Products, not SOTA on Amazon Reviews 2023, and not a general-SOTA claim of any kind.* The V1 VOID above stands unchanged as its own record (Appendix A.0).

5.3 The Musical_Instruments frequency-5 tail case (cross-dataset heterogeneity not established)

Beyond *whether* text helps overall (modestly: +2.7% on Video_Games, §5.1), we ask *where this additive frozen-text construction helps* (only early additive fusion of frozen text features is tested — §2.3 scope note). `evaluate()` reports by_popularity NDCG@10 over nominal train-frequency terciles {tail, mid, head} (bucketed by TRAIN frequency only, so no validation/test frequencies enter the bucketing; two cohort-definition defects are disclosed below), and we compare the full text stack against an ID-only ablation on the **same split and seed numbers**. The cold-start intuition — text should rescue rare items where ID embeddings are starved — holds on *one* of three datasets, and the MI difference survives an independent-arm analysis.

Randomization disclosure. Same-numbered text and ID arms are not initialization-paired: treatment-specific modules consume RNG before the shared backbone is constructed, changing backbone initialization and subsequent stochastic streams. Table 1d’s per-seed differences are therefore descriptive, all retained inference is independent-arm Welch/Satterthwaite, and the former paired CI/MDE/TOST analyses are withdrawn.

Table 1d: text – ID tail-tercile NDCG@10 contrast (same-seed-number arms — NOT initialization-paired; per-dataset estimates — cross-dataset heterogeneity not established, point 3).

dataset	catalog den- sity	tail Δ NDCG@10 (5-seed)	seeds posi- tive	verdict
Musical_Instruments	sparse	+0.000335 $\pm$ 0.000195 , Welch 95% CI [+0.000109, +0.000562] excl 0	5/5	frequency-5 tail advantage (independent-arm; §5.3)
Video_Games	dense	-0.000148 $\pm$ 0.000179	2/5	null — equivalence NOT estab- lished (retraction, point 2)
Beauty_and_PC	dense	-0.0000078 (3-seed, sample-sd 0.000020)	1/3	exploratory (3-seed; mixed eval geometry)

The historical MI contrast survives independent-arm analysis (Welch $t = 3.94$, $p = 0.014$, 95% CI [+0.000109, +0.000562]) and has tail HR $\Delta = +0.00109 \pm 0.00065$, but its absolute scale is small (≈ 29.6 versus 20.0 hits/seed); MI’s largest absolute text lift is at the head (+0.00527). On Video_Games, no tail difference is detected (Welch $p \approx 0.22$), but equivalence is **not** established. Beauty remains exploratory because one arm pair uses unmatched evaluation geometry. Most importantly, the repaired TFFV2 MI-VG contrast is +0.000247 ($p = 0.13$), so cross-dataset heterogeneity did not replicate and is not claimed (Fig. 3).

Why Table 1d is historical. Its nominal terciles mix zero-train-exposure targets with positive-frequency targets and split boundary-frequency ties by item-ID order; the original runs also lack per-row outcome sidecars. These arm-independent defects do not create a treatment-specific cohort, but they prevent the intended rare-item interpretation. TFFV2 therefore separates zero exposure, uses whole-frequency-group cohorts, and releases per-user outcomes.

Repaired-estimand campaign (TFFV2; 8 independent seeds/arm). On the tie-safe MI positive-frequency tail, text – ID is **+0.000420**, 95% CI [**+0.000181**, **+0.000660**], Welch $p = 0.0022$, passing the frozen Holm rule. The [1,6] band (+0.000546, $p = 0.0039$) and HR@10 (+0.001150, $p = 4 \times 10^{-5}$) agree. However, zero-exposure targets produce no hits through rank 100 in any of four categories, the exclude-frequency-5 sensitivity is nonsignificant (+0.000071, $p = 0.52$), and the frequency-5 group alone is +0.001379. We therefore report a **frequency-5-heavy positive-tail case**, not cold-start capability or a smooth rare-item law. VG is nonsignificant (+0.000173, $p = 0.14$), as is MI-VG (+0.000247, $p = 0.13$).

Evidence status. TFFV2’s endpoints were Git-frozen before launch, but interim test values became visible, its first independently verifiable timestamp postdates the first result, and the adjudicator was committed after some outputs existed. The eight-seed size was resource-based rather than prospectively powered, and the run manifests record concurrent tracked-document edits. TFFV2 is therefore **outcome-visible, not confirmatory**; its released sidecars and frozen estimator support reproducible estimates, not a claim of blinded confirmation. The repository retains the complete chronology, cohort files, logs, hashes, and errata.

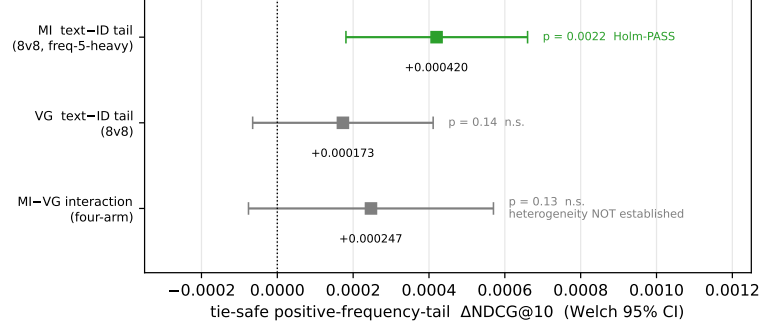
5.4 Titration: thinning tracks the head effect but not the tail effect (level contrasts under a bundled intervention; head trend exploratory and uncorrected; tail inconclusive)

We thinned Video_Games interactions over six rungs down to MI’s global density ($\rho=0.66$), holding n_{eval} at 94,762 and re-estimating text–ID differences with five seeds per rung. The two strata respond differently:

- **Head:** Δ rises overall from +0.00221 to +0.00354; all rungs are 5/5 positive. Spearman ρ_s versus density is -0.94 (exact $p=0.017$), with one adjacent reversal; the HR trend is weaker ($\rho_s=-0.71$, $p \approx 0.14$). This is an exploratory uncorrected rank trend on one fixed thinning draw.

The MI frequency-5 tail case and its two-axis descriptive contrast
 (A: TFFV2, 8 independent seeds/arm, outcome-visible; B-C: one fixed subset draw, 5-seed summaries, no draw uncertainty; heterogeneity not established, $p = 0.13$; AR2023 5-core LLOO, NDCG@10)

(A) TFFV2 repaired-estimand tail contrasts (one estimand; outcome-visible campaign)
 provenance: TFFV2_ADJUDICATION.md; cohorts tfv2_cohorts_*.json; data CSV alongside this figure



(B) Interaction-density titration
 (head tracks thinning / tail does not -- level contrasts)

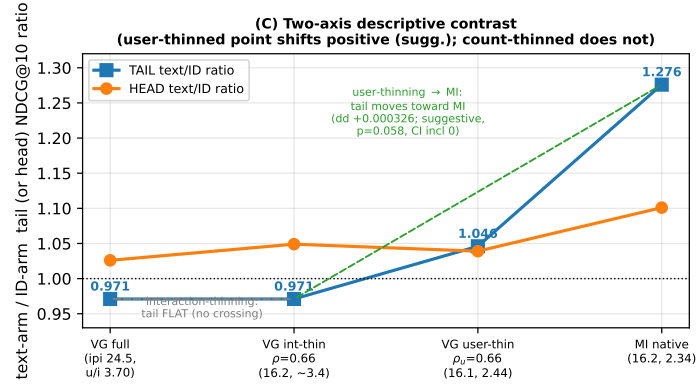
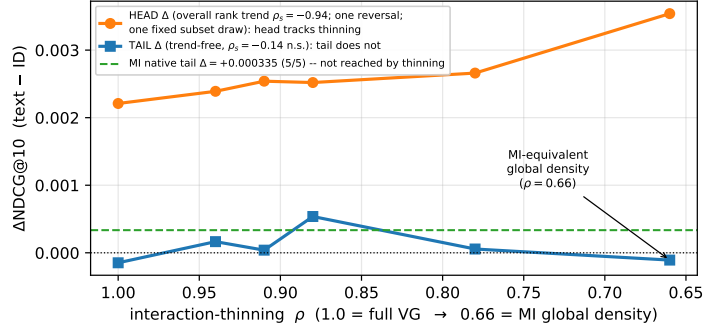


Fig. 3. (A) TFFV2 repaired-estimand tail contrasts — MI, VG, and the MI-VG interaction as Welch 95% CIs (8 fresh seeds per arm; one estimand; outcome-visible campaign, §5.3; cross-dataset heterogeneity not established, $p = 0.13$). The historical defective-cohort Table 1d values are tabulated, not plotted. (B) Interaction-density titration level contrasts and (C) two-axis descriptive ratios — both on ONE fixed subset draw with no draw/seed uncertainty shown; arms are not initialization-paired (§5.3); the panel-C shift is suggestive only ($p = 0.058$, CI includes 0). Plotted values, estimator, and analysis IDs: figures/fig_tail_law_mechanism_data.csv.

- **Tail:** Δ is non-monotone ($\rho_s = -0.14$). At MI-matched density it is -0.000108 ± 0.000503 (1/5 positive), so thinning does not reproduce MI’s $+0.000335$ tail estimate. Without a prospective equivalence margin this is inconclusive, not evidence of no effect.

The full ladder is Table S2. The head trend and tail non-reproduction are level contrasts under a bundled intervention; their interaction is nonsignificant. Global density is therefore not identified as the tail mechanism.

5.4.1 The level contrasts in one scale-free table: the cross-dataset tail/head arm ratio (descriptive; no interaction-test support). To separate absolute scale from arm contrast, Table 5 reports the text/ID NDCG ratio at full VG density, VG thinned to MI density, and native MI. Ratios above one favor text.

regime (interactions/item)	n (text/id)	TAIL text	TAIL id	TAIL ratio	tail Δ	HEAD ratio
VG full density (ipi 24.5)	5 / 5	0.004919	0.005068	0.971 (−2.9%)	−0.000148	1.026 (+2.6%)
VG thinned → MI-density ($\rho=0.66$, ipi 16.2)	5 / 5	0.003569	0.003677	0.971 (−2.9%)	−0.000108	1.049 (+4.9%)
MI native (ipi 16.2)	5 / 5	0.001551	0.001216	1.276 (+27.6%)	+0.000335	1.101 (+10.1%)

The head ratio rises from 1.026 to 1.049 as VG is thinned, toward MI’s 1.101. The tail ratio remains 0.971 at both VG densities, far below MI’s 1.276. Thus matching global density does not reproduce the MI tail advantage on this draw. The earlier two-seed “trough” interpretation is retracted; the five-seed endpoint is near zero with a wide interval. This remains a level contrast, not a significant stratum-by-density interaction.

One hypothesis is that user connectivity or item-text quality differs between native MI sparsity and artificially thinned VG. Neither is identified here: user thinning changes several dataset properties at once, and an item-text permutation control has not been run.

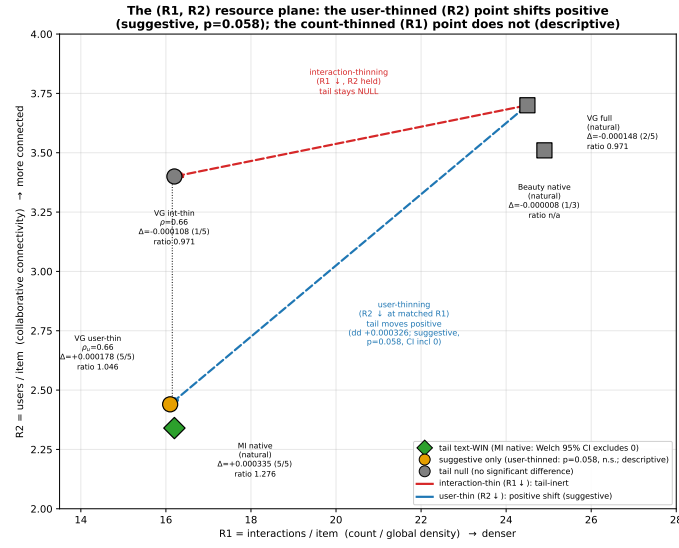
5.4.2 User-mode titration: a suggestive tail-level response at MI-matched connectivity (bundled intervention; descriptive). User thinning preserves each retained user’s history while lowering users/item. At $\rho_{\text{user}}=0.66$ it matches the interaction-thinned rung’s interactions/item (16.1 versus 16.2) and approaches MI’s users/item (2.44 versus 2.34). Because user composition, degrees, update counts, and topology also change, this is a bundled-scheme contrast, not single-factor identification.

The read comes out on the predicted branch — a positive shift (suggestive):

regime (int/item, users/item)	TAIL ratio	tail Δ (5-seed)	HEAD ratio
VG full (24.5, 3.70)	0.971	−0.000148	1.026
VG interaction-thinned $\rho=0.66$ (16.2, ~3.4)	0.971	−0.000108 $\pm$ 0.000503 (1/5 pos)	1.049
VG user-thinned $\rho_{\text{user}}=0.66$ (16.1, 2.44)	1.046	+0.000178 $\pm$ 0.000153 (5/5 pos, $t \approx 2.6$)	1.039
MI native (16.2, 2.34)	1.276	+0.000335 $\pm$ 0.000195 (5/5 pos)	1.101

The tail ratio moves from 0.971 to 1.046. The independent-arm difference-in-differences is $+0.000326$ ($p=0.058$, 95% CI $[-0.00001, 0.00067]$); the within-rung tail level is $+0.000178$ ($p \approx 0.16$). The former paired analysis is retracted because arms are not initialization-paired. Fig. 2 therefore shows a suggestive direction, not a confirmed effect or connectivity attribution.

Lower user-thinning rungs do not support a “peak then collapse” interpretation: $\rho_{\text{user}}=0.50$ is $+0.000236 \pm 0.000532$, while $\rho_{\text{user}}=0.40$ exhibits a text-arm floor collapse. The user-thinned estimate also remains below native MI. Overall, interaction thinning tracks the head level, while user thinning gives only a suggestive tail-level shift. Shared intervention draws omit subset uncertainty, the stratum-by-density interaction is nonsignificant, and the mechanism remains



AR2023 5-core LLOO, full-catalog NDCG@10; VG runs 5-seed best-by-val ($n_{\text{eval}}=94,762$, $\text{tail}_n=10,900$ frozen), Beauty 2-3 seed. The two VG thinned points share $R1 = 16.2$ (matched count) but differ in $R2$; only the $R2$ -thinned point moves to a positive tail point estimate beside MI (suggestive).

Fig. 4. The ($R1 = \text{interactions/item}$, $R2 = \text{users/item}$) resource plane — the two VG thinned rungs share $R1$, but only the user-thinned ($R2$) point moves to a positive tail point estimate beside MI native (suggestive, $p = 0.058$; descriptive contrast; bundled interventions on one fixed subset draw; the §5.4.2 dd inference is suggestive, not significant, under the corrected independent-arm analysis).

unidentified. A prior spectral explanation is retracted because it did not measure the claimed spectrum [Baik et al. 2005; Gavish and Donoho 2014; Marchenko and Pastur 1967].

Provenance: released results_USERTITR_*.json; five seeds per rung; $n_{\text{eval}}=94,762$ and $\text{tail}_n=10,900$ fixed. Arms share seed numbers but are not initialization-paired. All model seeds use one intervention draw.

5.5 Screening log of capacity-adding probes (a search record, not powered exclusions)

A central empirical finding of this work is **asymmetric**: the only levers that moved test NDCG@10 were *capacity-restricting* (label smoothing, the causal FIR filter) or classical inductive biases already in Table 1 (the time bias); **no capacity-adding probe we tried showed a benefit at its tested power (mostly single-seed)**. We document the full set here as a first-class result — negatives are part of the contribution, and several are *interpretable* negatives in which a learnable scalar, free to engage the mechanism, was driven by the optimizer to (or past) zero. This is observational evidence — most rows are single-seed probes (power note below) — that the in-environment ceiling was not left unexplored on these axes; the residual gap to the published HSTU-BLaIR 0.0760 could not be isolated (the pinned reference environment is not installable here, and the shimmed local execution of §5.6 did not include the Video_Games configuration); what the probe set shows is that none of the tested probes (mostly single-seed; Table S1 power note) produced a gain — observations that leave the gap unexplained, not exclusions.

All entries are best-by-val test NDCG@10, full-catalog eval ($n_{\text{eval}} = 94,762$ on Video_Games), on the same fixed protocol. Δ is versus the stated base; for the three learnable-scalar probes (W1/X1/Y1) we report the absolute single-seed test next to its base band, since the headline test is statistically indistinguishable from the base and the *learned*

scalar is the load-bearing readout. Several rows are single-seed — sufficient for a negative-result map (a probe that fails to clear its base on one seed, or whose own learned scalar switches it off, does not warrant five seeds).

Unequal power, stated plainly. The probes in Table S1 do *not* carry equal evidential weight: most rows are single-seed exploratory probes, while a few were run at full multi-seed power. Only the multi-seed, pre-declared negatives — the conn-gate five-seed confirmation (cadence-caveated; see the row note) and the titration nulls of §5.3–§5.4 (the VG tail null — former equivalence claims retracted, §5.3 — and the 5-seed-locked ladder rungs) — were run at full pre-declared power (no null is thereby confirmed); the **Beauty near-zero tail estimate is a 3-seed exploratory observation** (mixed evaluation geometry, §5.3) and carries **no** confirmatory weight. The remaining rows are exploratory documentation of the search space, recorded for completeness and interpretability, not powered hypothesis tests. Full per-probe seed counts are given in the table’s *n* column.

The full screening log — all 18 capacity-adding / alternative-mechanism probes with per-probe Δ NDCG@10, seed counts, and learned-scalar readouts — is **Table S1** (Supplement §S.4); it is summarised above and analysed next.

The pattern is consistent and is itself the finding: across loss-side (W1, Y1, sampled-softmax), embedding-side (X1, GD1, text-init), attention-side (c3, n_heads, text-sim), encoder-side (BLaIR, dual-text), and auxiliary-objective (c1, c2, CL4SRec, EMA) axes, no capacity-adding probe produced a test gain at its tested power (mostly single-seed; power note above), and three free scalars (W1’s β , X1’s c , Y1’s T) were each switched off — or, for W1, reversed — by the optimizer. Only anti-overconfidence regularizers (label smoothing) and the restricted causal-filter inductive bias (Table 1) transferred. These are hypothesis-generating observations at their stated (mostly single-seed) power, not powered exclusions; **we do not claim they bound an in-environment ceiling**. (An earlier spectral-irreducibility clause here is retracted — see the §5.4.2 retraction note.) The conn-gate row deserves separate emphasis: it is the only entry tested at full five-seed power and the only one that directly operationalizes *this paper’s own intervention-associated tail pattern* — the collaborative-connectivity association of §5.4.2. A single seed showed a $+0.00053$ tail lift, but the pre-declared five-seed confirmation collapsed to $+0.0001 \pm 0.0003$ (95% CI $[-0.00025, +0.00045]$, includes zero), and the free gate scalar α was driven *below* its initialization (voted off; the learned value is recorded in 4/5 seed logs — the fifth predates the log line, and the graph reports the recovered count). **Cadence caveat (disclosed 2026-07-19):** the conn-gate arms were evaluated only at epochs {10, 20} while their base evaluated every epoch, so best-by-val checkpoint opportunities were unequal across arms; the tail CI-includes-zero verdict is unchanged by this, but the small overall delta is cadence-sensitive and we rest no claim on its sign. This is the cleanest illustration of the paper’s central distinction: a mechanism-associated pattern can be *supported* by controlled thinning contrasts (the matched-R1 level contrast) yet not be *actionable* through a learnable cold-start gate — one *interpretation* — a hypothesis, not an identified mechanism — is that the connectivity signal the model needs is already latent in the frozen text features, leaving no residual for an explicit gate to exploit.

5.6 Executing the reference implementation locally (post-hoc comparator regeneration)

Late in this work we succeeded in running the **reference implementation itself** — the HSTU-BLaIR repository at its released commit, unmodified (clean git tree): its preprocessing, its gin configurations, its trainer, and its eval protocol — end-to-end on our hardware. The pinned environment (torch 2.2.2 / fbgemm_gpu 0.6.0) remains uninstalleable here (no sm_120 kernel images, no Windows wheels); what makes execution possible under our unpinned torch 2.11 environment is that the repository’s research path uses exactly **three** fbgemm operators, all pure data movement (asynchronous_complete_cumsum, jagged_to_padded_dense, dense_to_jagged), which we re-implement in

plain PyTorch with self-tested forward/gradient parity, plus a world-size-1 identity DDP wrapper and logging/data-acquisition patches (full shim inventory, per-run provenance, and artifacts: `THEIRS_ON_OURS_REPORT.md`). No model math is touched — the layer norms, fused projections, SiLU pointwise attention, biases, loss, and scoring all execute in the reference repository’s own unmodified lines, consistent with the core-block parity evidence of §3.2. Their own pre-processing pipeline ran with its dataset-size assertions left active and **passing** on both corpora (Musical_Instruments: 24,587 items / 57,439 users; Office_Products: 77,551 items / 223,308 users — the comparator paper’s exact statistics).

Two single-run results (their configurations as shipped, 101 epochs; our RTX 5060 Ti):

run (their code, their data, their eval)	published	local, final epoch	local, best full eval
MI HSTU-BLaIR — NDCG@10	0.0406	0.0391	0.0406 (epoch 35 — exact)
MI HSTU-BLaIR — HR@10	0.0733	0.0716	0.0743
MI HSTU-BLaIR — MRR	0.0371	0.0355	0.0369
Office SASRec — NDCG@10	0.0153	0.0174 (+13.9%)	0.0177
Office HSTU-BLaIR — NDCG@10	0.0271	0.0275 (+1.6%)	0.0279

The Musical_Instruments comparator regenerates locally. The final-epoch full-corpus eval lands within 2.3–4.3% of every published metric above, and the best full-eval epoch hits the published NDCG@10 exactly; the run oscillates in a 0.0384–0.0406 band from roughly epoch 30 onward (the comparator’s own README notes run-to-run variability). The published 0.0406 that §5.2’s pre-declared confirmation is measured against is therefore no longer only a transcribed constant: a local run of the generating code regenerates it under the unpinned shimmed research path, and our pre-declared CI lower bounds (0.04096 / 0.04083) sit above both its best (0.0406) and final (0.0391) local readings. **The Office_Products floor anomaly resolves:** their own SASRec configuration lands +13.9% above its published row here, crossing the published value at epoch 20 of 101 and never re-entering — which, combined with our floor run, decomposes the +44% floor-check failure entirely into published-row conservatism and baseline-strength protocol differences, leaving nothing attributable to our split or eval (resolution in Appendix A.0; the VOID is deliberately retained).

Caveats (why these are regenerations, not reproductions). Single runs (the reference code seeds only Python’s random, so even in place it is not run-to-run deterministic); an unpinned environment (torch 2.11 vs pinned 2.2.2, transformers 5.7 vs 4.51, different cuBLAS/TF32 kernel versions); a different GPU than the comparator used; and the shims themselves — though the shim risk has since been closed on CPU: under the pinned torch 2.2.2 itself, the real `fbgemm-gpu(-cpu)==0.6.0` operators match our shims **bit-exactly (max abs diff 0.0, forward and gradient)** on every input the pinned binary accepts, the reference block under {pinned torch + real fbgemm} vs {pinned torch + shims} is bit-exact at every stage, and the remaining torch 2.2.2→2.11 block-level numerics drift is bounded at 4.8×10^{-7} **per stage** / 2.4×10^{-7} **end-to-end**, with negative controls confirming the harness detects 1-ulp perturbations (`PINNED_ENV_PARITY_REPORT.md`); the pinned stack’s GPU (cu121) kernel numerics remain the one locally untestable component. We therefore cite these as **environment-caveated single-run regenerations** and change no claim wording on their basis: §5.2’s claim remains a point-estimate comparison against the published number.

5.7 Pre-declared fresh-seed hybrid: late fusion with a closed-form item-item model (E-F; adjudicated 2026-07-23; W-H-POS on all three categories)

Provenance and system. An author-operated side campaign (2026-07-22; outcome-visible development on historical seeds, disclosed as such) found that late score fusion of the frozen sequential stack with EASE [Steck 2019] — a closed-form linear item-item model fit on the *train split only* — improved MI test NDCG@10 on every historical seed. PREREG_HYBRID_V1.md (frozen and committed before launch, with its mechanical adjudicator) is the fresh-seed replication: per user, fused = $z(\text{seq}) + w \cdot z(\text{EASE})$; (l2, w) selected on *validation only* from frozen grids; the fusion stage evaluates test once, at the selected pair (the underlying trainer additionally logs test curves at eval epochs under the pipeline’s long-standing best-by-val protocol — a standing test exposure disclosed in §5.3 process disclosure (i), not a fusion-stage selection input); Completed per-seed fusion results were also read for operational monitoring before the full family finished; PREREG_HYBRID_V1 registered no no-interim clause, and the once-only adjudication ran after all files existed (disclosed). Five fresh seeds 20260721–25 per category; integrity gates all passed (per-seed reconstruction fidelity <0.0005 vs the recorded base result; frozen-grid membership; canonical n_eval; mechanical config equality to the per-category frozen reference), adjudicator exit 0.

Result (frozen wording; ‘pre-declared’ replaces the prereg’s synonym for the manuscript’s uniform terminology). Under the pre-declared fresh-seed replication (PREREG_HYBRID_V1, 5 seeds), late z-score fusion with a train-only EASE model changed test NDCG@10 by **+0.00244** [**+0.00219, +0.00268**] on Musical_Instruments, **+0.00261** [**+0.00211, +0.00310**] on Industrial_and_Scientific, and **+0.00317** [**+0.00286, +0.00348**] on Video_Games These are ordinary paired 95% CIs; raw paired p-values are 1.0e-5 on MI, 1.3e-4 on IS, and 9.0e-6 on VG, all significant after Holm adjustment. Holm adjusts p-values and decisions, not the intervals; significance attaches only to fused-vs-sequential contrasts. The hybrid is a two-scorer system; published comparator numbers are single-scorer, single-run values, and every comparison below remains a point-estimate comparison under the environment caveats of §3. Fresh-seed sequential-only means for context: MI 0.04155, IS 0.03340, VG 0.06713.

Published-comparator rows (frozen wordings). The fused system’s five-fresh-seed mean test NDCG@10 on Musical_Instruments, **0.04399**, exceeds the published single-run HSTU-BLaIR point estimate 0.0406 under our environment-caveated regeneration protocol (point-estimate comparison; no distributional claim). On Video_Games the fused five-fresh-seed mean, 0.07031, remains below the published single-run 0.0760. A five-checkpoint ensemble of the same frozen stack, fused with EASE under the same validation-only rule, scored **0.04557** on Musical_Instruments (single evaluation; ensemble frame — comparable only to other ensembles).

Attribution and scope. EASE is Steck (2019); z-score late fusion is standard. The local contribution is empirical: the measured complementarity between a text-augmented sequential scorer and a closed-form co-occurrence scorer under the 5-core LLOO full-catalog protocol, quantified under a frozen plan on fresh seeds. EASE alone is far weaker than the sequential model on these catalogs (side-campaign observation; the gain is complementarity, not a stronger component). The dense closed-form fit is infeasible at Office/CDs/Beauty catalog sizes; that scope exclusion was declared in the prereg, and a content-based third scorer that avoids the item-item inversion was subsequently studied at five-category scale (§5.8; outcome-visible, protocol-deviated — see its classification). Artifacts: results_{MI,IS,VG}_HYBRIDV1_base_seed*.json (+.fusion.json, per-user records), results_MI_HYBRIDV1_ensemble5.json, hybrid_v1_adjudication.json.

5.8 Sparse-warm text-fusion study on all five categories (E-G; outcome-visible, protocol-deviated; descriptive estimates, no pre-declared-confirmation status; a near-threshold redistribution, not cold start)

Classification. All 25 runs completed, but the campaign is outcome-visible and protocol-deviated. Interim endpoints were exposed, the literal frozen configuration gate fails on MI and VG, and the amended gate postdates outcome visibility. The literal failure governs; the amended PASS is a sensitivity only. A planned replication was also exposed before adjudication. Accordingly, every estimate below is descriptive and conditional on optimizer seeds over one repeatedly used split.

System. At evaluation time, a history-centroid text score is added to the sequential score and, where feasible, EASE. Text weights depend only on train-frequency bins and are selected on validation under an overall non-inferiority constraint. The text scorer fits no parameters, but the sequential and EASE components are trained or fit; this is not a training-free recommender.

The repository retains the as-launched gate, later sensitivity gates, their hashes, and the exact chronology. This preserves the defect without converting a post-outcome repair into a frozen decision.

Descriptive result. Tail-bin NDCG@10 changes were +0.00217 [0.00196, 0.00239] on MI, +0.00243 [0.00218, 0.00267] on IS, +0.00337 [0.00305, 0.00370] on VG, +0.00114 [0.00105, 0.00123] on Office, and +0.00481 [0.00472, 0.00490] on CDs. Overall changes were near zero. These intervals are retained as descriptive arithmetic, not as pre-declared confirmation.

The gain is entirely frequency 1–5; frequency-zero targets never change at rank 10. Mid and head means decline in every category, so this is a sparse-warm redistribution rather than free overall accuracy. All validation choices hit the text-weight grid ceiling, leaving the optimum unbracketed. With five seeds, the exact two-sided sign-test floor is 0.0625.

Evaluation-time content fusion is established prior art [Brody and Lagziel 2024; Collins et al. 2025; Lichtenberg et al. 2025; Liu et al. 2023b; Wang et al. 2024, 2025; Xu et al. 2024]. We claim no architectural novelty for this element; it is retained only as a protocol-deviated redistribution study on fixed splits. Full operational chronology and artifacts are in the versioned supplement and repository.

6 Discussion

6.1 Interpretation

The main result is deliberately modular and narrowly identified. Positive FIR–identity contrasts on three outcome-visible Amazon categories are joined by a positive, thresholded Software robustness contrast. They show improvement only in those tested Amazon settings: the prospectively frozen MovieLens 1M study did not replicate learned FIR against identity or pointwise. V3 improves the protocol chronology, but unverified non-visibility of earlier V2 validation output, same investigator/code lineage/Amazon family, and local custody keep it outcome-known and non-independent. An equal-parameter current-position-only DCT/GELU placebo does not show a detected improvement over identity, and learned FIR exceeds it under matched initialization. This discriminates learned FIR from that compound residual but does not isolate temporal access because basis/rank, activation, channel mixing, and temporal access change together. The shared causal filter is also not separated from learned per-channel taps. On MovieLens, shared, grouped, and low-rank FIR arms met the noninferiority margin against learned FIR, but the learned-FIR effect gate failed; parsimony is therefore conditional rather than evidence of useful compression. The official-code WEARec feasibility run also scored below our existing full-model reference under the shared evaluator. The frozen

AlphaFuse-style package scored above its repository SASRec-ID backbone but below that same full-model reference. Together these outcomes improve current-baseline coverage while preserving the mixed ranking. They are not equal-architecture/capacity/tuning comparisons, do not isolate null-space fusion, and cannot supply independent confirmation. The evidence establishes neither per-channel-tap necessity, a universal frequency-filter benefit, cross-domain generalization, nor a new end-to-end architecture.

The supporting text studies sharpen that boundary. Frozen text features add only +2.7% overall on Video_Games, and TAPE is sub-additive (+0.0009 in one seed; +0.0004 in the four-seed check). The repaired MI analysis finds a small frequency-5-heavy benefit rather than cold-start capability: zero-exposure targets receive no hits through rank 100, the exclude-boundary sensitivity is nonsignificant, and the MI-VG contrast does not replicate. In the Beauty_and_PC scan (Appendix A.1), the positive rows bundle BLAIR, richer text, and the MLP adaptor, while the single MiniLM Linear-to-MLP comparison shows no improvement. The scan therefore does not isolate whether encoder, content, projection, or their interactions account for the bundled difference.

6.2 Practical implications

The FIR residual is small, causal, and detachable: it can be initialized as the identity and added without changing the item scorer or evaluation protocol. MovieLens resource measurements show no practically meaningful accuracy-resource frontier for learned FIR in the tested setting; latency and peak-memory differences are small, hardware-specific, and subordinate to the failed effect gate. In this implementation, chunked full softmax makes 207k-item training feasible, cached item features substantially reduce evaluation cost, and right-padding with a causal-only mask avoids the NaN failure observed with the former masking path. Conversely, heavy dropout, removal of learned item embeddings, and simple subsequence augmentation did not transfer in the Beauty supporting study; these are observations under this pipeline, not general exclusions.

6.3 Claim limits and remaining tests

- **Comparator uncertainty.** The counted Musical_Instruments and Office V3 comparisons are against published single-run point estimates and an environment-matched single-run regeneration; no paired or distributional superiority over HSTU-BLAI is claimed. The official end-to-end CUDA/Triton system cannot run on the local sm_120 GPU. Only the research block at an explicitly aligned configuration is numerically matched; trained-checkpoint and end-to-end equivalence are not claimed.
- **Historical, canonical, control, and frozen Software evidence.** Same-numbered historical arms were not initialization-paired, so their paired interpretations are withdrawn. E-A uses its frozen Welch/Satterthwaite analysis; canonical breadth uses matched configuration blocks, but both reuse outcome-visible splits. The six-arm final evaluation was sealed, but ran across a dirty evolving tree and lacked independent sidecar custody. Fixed MA/HP are algebraically redundant; shared and nonlinear controls prevent a learned-tap-specific claim. The pointwise study adds a learned-FIR-versus-compound-current-only contrast on outcome-known MI, not temporal isolation or per-channel necessity. Software V3 used validation-only selection and sealed final evaluation, but V2-output non-visibility is not independently established; we classify it outcome-known/exploratory. Its local seals are hash-linked rather than immutable external custody, and its frozen tag has a disclosed raw-line-ending reference-hash replay defect. Office V1 remains **VOID**; Office V3 supports only its frozen point-estimate wording.

- **Prospective MovieLens boundary.** The non-Amazon study improves selection chronology and TEST sequestration, but it is same-investigator and same-code-lineage on one dataset and one global-time split. Its negative verdict blocks a cross-domain FIR-benefit claim. Noninferiority of the parsimonious arms is conditional on the failed learned-FIR replication gate; it is not equivalence to identity, a deployment utility claim, or independent confirmation. User/item bootstrap intervals are fixed-dataset sensitivities. The public graph can recompute the released aggregate seed vectors but cannot independently replay non-redistributable MovieLens records or private endpoint files.
- **WEARec current-baseline boundary.** The official model/training code was run under the paper split, complete-history mask, full-catalog evaluator, cutoff, and tie rule after validation-only preset selection. The resulting WEAREC-BELOW-EXISTING-REFERENCE verdict is same-investigator evidence on an outcome-known split. Architecture, loss, schedule, and tuning budgets differ; the unpaired Welch contrast is descriptive, and neither independent confirmation nor a SOTA claim follows. The graph replays NDCG vector arithmetic, not private endpoint extraction or unreleased HR/MRR vectors.
- **AlphaFuse-style current-comparator boundary.** E-E V3 froze the official upstream representation and repository SASRec classes, shared training configuration, complete-history-masked evaluator, and eight fresh optimizer seeds per arm before launch. MiniLM-384 replaces the published AlphaFuse text vectors. The positive package-minus-ID interval is countable current-comparator evidence, but same-numbered seeds are not paired and the arms differ in text availability, initialization, trainable capacity, parameter allocation, and architecture. The negative package-minus-existing-reference interval is reported with equal prominence. Neither contrast isolates null-space fusion, reproduces a published AlphaFuse table, establishes equal tuning, supplies independent confirmation, or permits a SOTA claim. User/item bootstrap intervals are fixed-dataset sensitivities; the public graph validates their recorded schema/intervals and endpoint hashes but cannot replay private endpoints or record-level resampling.
- **Tail and probe scope.** The tail result is AR2023-specific, small in absolute terms, concentrated at train frequency 5, and not replicated on a non-Amazon domain. The original cohorts mixed exposure regimes and split frequency ties; TFV2 repaired those estimands but was outcome-visible and is not classified as confirmatory. Most capacity probes are single-seed screens, and nonsignificance is never interpreted as equivalence.
- **Open controls.** Item-text permutation and objective/target-multiplicity parity controls remain unrun, so the small text-stack gain cannot yet be attributed entirely to semantic alignment. The thinning studies intervene synthetically on one fixed draw and do not identify the real-world data-generating process.
- **Provenance and input scope.** One confirmation campaign ran at a documentation-only descendant of its frozen commit; code hashes and empty protocol-file diffs support code identity, but the literal commit-equality deviation remains disclosed. The canonical model consumes only item IDs, timestamps, and frozen item-text embeddings—never outputs or representations from a comparator model.

7 Conclusion

This paper contributes a narrow causal-filter module and an evaluation discipline for testing it. The discipline combines git-frozen protocols, a fail-closed artifact graph, environment-caveated comparator regeneration, and symmetric adjudication. Its value is illustrated by the Office_Products V1 campaign: its own floor check VOIDed the comparison, and that verdict remains in the record even though a redesigned protocol later passed on fresh seeds.

The main empirical result is a family of identity-initialized, gradient-active, strictly left-causal residual modules. Learned FIR–identity estimates are positive on Musical_Instruments (+0.002265, 95% CI [0.001928, 0.002602]), Industrial_and_Scientific (+0.002110 [0.001820, 0.002399]), and CDs_and_Vinyl (+0.006150 [0.005849, 0.006450]), but these are outcome-visible internal estimates. A separately frozen Software robustness attempt found learned FIR minus identity +0.005062 [+0.004591, +0.005533], with 8/8 positive paired differences and the ordinary 95% CI lower bound above its pre-declared +0.000500 reporting threshold. Because earlier V2 validation-output non-visibility is not independently established and the work remains same-investigator, same-code-lineage, same-Amazon-family under local same-user custody, it is outcome-known robustness rather than independent confirmation. In the active-control study, every trainable left-causal arm improves identity; learned taps beat the redundant fixed-filter parameterizations but are not separated from the shared causal filter or parameter-matched nonlinear causal control. In the subsequent equal-parameter current-position-only placebo study, pointwise minus identity is -0.000069 [-0.000200, +0.000061], while learned FIR minus pointwise is +0.001941 [+0.001788, +0.002095]. This discriminates learned FIR from that compound current-only residual in outcome-known MI, but does not isolate temporal access. It does not establish per-channel-tap necessity. In the prospectively frozen MovieLens 1M study, however, learned FIR did not replicate against identity (+0.000000 [-0.000074, +0.000075], $p_{\text{Holm}} = .995$) or pointwise (+0.000035 [-0.000057, +0.000127], $p_{\text{Holm}} = .796$). Shared, grouped, and low-rank arms met the pre-declared noninferiority margin against learned FIR only conditionally: the learned-FIR effect gate failed. The evidence therefore supports neither unique necessity of learned per-channel FIR coefficients, a general FIR benefit, superiority over the stronger external reference, nor independent cross-category confirmation.

The official-code WEARec current-baseline campaign reached mean NDCG@10 0.059184 [0.058674, 0.059693], below the existing six-seed full-model reference 0.067337 [0.067063, 0.067611]; the descriptive unpaired contrast was -0.008154 [-0.008689, -0.007618]. This closes only the narrow official-code/equal-evaluation feasibility check. It does not equalize architectures, losses, schedules, or tuning budgets and is not independent confirmation or SOTA evidence.

The frozen AlphaFuse-style current-comparator campaign reached NDCG@10 0.048273 [0.048129, 0.048416], above its repository SASRec-ID backbone at 0.039024 [0.038106, 0.039941] by +0.009249 [+0.008329, +0.010169], but below the existing full-model reference by -0.019065 [-0.019347, -0.018783]. This countable mixed result supports the value of the tested whole representation package over that ID backbone; it does not isolate null-space fusion, equalize architecture/capacity/tuning, reproduce AlphaFuse’s published text setting, or provide independent confirmation or SOTA evidence.

The secondary result is narrower. Frozen text features improve the repaired Musical_Instruments frequency-tail endpoint (+0.000420 [0.000181, 0.000660]), but the gain concentrates at train frequency 5; removing that boundary group leaves no detected effect. The corresponding cross-dataset contrast is nonsignificant, and thinning interventions do not identify a tail mechanism. We therefore report a dataset-specific frequency-5 case, not a general cross-dataset result. TAPE, late fusion, sparse-warm redistribution, and the probe screen are supporting or descriptive studies, not additional headline contributions.

The evidence has important limits: uncertainty is over optimizer seeds on reused fixed splits; comparator uncertainty is not distributionally matched; several campaigns are outcome-visible or protocol-deviated; and no item-text permutation control, temporal split replication, or deployment study has been completed. The defensible contribution is thus modular rather than architectural: a small causal FIR realization, its measured internal effect under the stated protocol, and an auditable record that preserves nulls, deviations, and VOID outcomes alongside positive results.

8 Code and Data Availability

Code, frozen protocols, result artifacts, and build scripts are public at <https://github.com/Ray0419/bestrec-sota-results>. The tracked `RELEASE_MANIFEST.json` binds source code, split derivatives, text caches, result JSONs, released per-user sidecars, and the pinned HSTU-BLaIR submodule by SHA-256. Running `bootstrap_public_clone.py` fetches and verifies release-class assets; running `rebuild_hstu_submission.py --strict` then checks HSTU parity, recomputes the 199 artifact-gated cells, verifies the manifest, and executes every governed adjudicator.

The numerical graph and release manifest have different scopes: every declared empirical cell is graph-bound, but not every transitive graph source is separately enumerated in the release manifest. Table 0’s literature attribution remains citation-checked authored prose, but its quantitative FIR fields are generated from active graph cells and the strict rebuild verifies the generated region. Table 0 is not a separate numerical graph family. We make no broader “every printed claim is hash-manifested” claim.

MovieLens 1M record rows, transformed splits, checkpoints, endpoint files, and per-user sidecars are not redistributed under the ML-1M README. The public release contains the frozen acquisition/training/evaluation code, source and split hashes, aggregate seed vectors/statistics, and adjudication. Consequently, the public graph recomputes the aggregate MovieLens claims but cannot independently replay the private record-level endpoint extraction.

The WEARec release contains the frozen official-code adapter, validation-only selection record, eight-seed aggregate adjudication, reference-vector identities, and private endpoint hashes. Its graph cell independently recomputes the released NDCG summaries and descriptive Welch arithmetic. Checkpoints, sealed endpoint files, and per-user sidecars remain private, and HR/MRR raw-vector arithmetic is therefore not publicly replayed.

The public `v0.9-audit-evidence` release is the working evidence store. Historical campaigns for which checkpoints or per-user records were not retained are labeled as such and require regeneration for rank-level reconstruction. Raw Amazon Reviews 2023 archives are not redistributed; derived splits are tied to their public source and preprocessing code by hashes. Local-only audit sidecars are outside every printed claim and are precommitted by digest for reviewer access. A new archival deposit will be cut only after the manuscript, author metadata, licenses, and clean-clone build are finalized; the older `v1.1.11-deposit` tag is retained as a historical boundary, not represented as current.

9 Acknowledgments

We thank the Amazon Reviews 2023 maintainers [Hou et al. 2024] for releasing the dataset and accompanying reference implementations. The MLP adaptor design used in our best variant is from their published SASRecText configuration.

10 Ethics and Data Governance

This work uses only the public, pseudonymized **Amazon Reviews 2023** dataset (Hou et al., 2024; McAuley Lab), obtained from its official distribution. The dataset’s maintainers state publicly that they are not in a position to assign a license or dictate usage terms; **that statement is not an affirmative permission grant, and we do not treat it as one**. We redistribute derived interaction-split CSVs (never the raw dataset) with attribution on the basis of the dataset’s public research availability, we will remove them immediately on maintainer, platform, or venue request, and we flag the redistribution basis for venue-level review rather than asserting a legal right. Scope of data actually consumed: interaction tuples (pseudonymous user identifiers — remapped to dense integers by our preprocessing — item identifiers, and timestamps) and **item metadata text** (titles, categories, and store/seller name — the AR2023 store field; an earlier label said “brand”, corrected 2026-07-20: the frozen cache text template’s literal brand: prefix is

retained as-is inside the cached strings and documented as a misnomer, because regenerating the caches would invalidate every frozen-text result) for the frozen text encoders. We do **not** process review text bodies, ratings-as-text, or product images, and we make no attempt at re-identification of any user. Released artifacts follow the same boundary: per-user evaluation sidecars contain only remapped integer IDs, target item IDs, and rank outcomes; the raw dataset is **not redistributed** (our releases carry SHA256 hashes and regeneration scripts instead, `RELEASE_MANIFEST.json`). No new data was collected and no interaction with human subjects took place; no institutional review determination was sought or is claimed — the data are public, platform-pseudonymized product reviews, and we state that as a fact rather than adjudicating any jurisdiction’s requirements; ACM policy places responsibility for applicable institutional, ethical, and legal compliance on authors, and we accept it, with this section’s factual basis available as supporting documentation on request. **Sidecar identifier surface, stated exactly:** the released per-user sidecars carry our preprocessing’s dense remapped user indices (`user_id` 0, 1, 2, ...), a target-item index, and rank outcomes — not hashes; those dense indices are deterministically linkable to the platform’s pseudonymous identifiers through the released split CSVs, so the sidecars are exactly as pseudonymous as the public dataset itself, no more — we state this linkage surface plainly instead of calling the sidecars anonymous. Released derivatives are retained under the tagged GitHub release; removal requests (maintainer, platform, or a data subject acting through the platform’s pseudonymization) will be honored by deleting the affected assets and regenerating the release manifest. Finally, this is an offline evaluation study: recommender systems deployed on such data can amplify popularity and exposure biases — our long-tail analyses quantify one aspect of that concern — and nothing in this paper constitutes a deployment claim.

References

- Soohan Abbasi, Shahid Munir Shah, Rafia Shaikh, and Mahmoud Aljawarneh. 2026. SISA-Rec: A Semantically Integrated Sequential Recommender with Contrastive Alignment. *arXiv:2607.11168* Concurrent preprint; Amazon Reviews 2014 Beauty/Toys, not the paper’s AR2023 protocol; cited for modular semantic-integration coverage only.
- V. W. Anelli, A. Bellogín, A. Ferrara, D. Malitesta, F. A. Merra, C. Pomo, F. M. Donini, and T. Di Noia. 2021. Elliot: A Comprehensive and Rigorous Framework for Reproducible Recommender Systems Evaluation. In *SIGIR*.
- J. Baik, G. Ben Arous, and S. Péché. 2005. Phase Transition of the Largest Eigenvalue for Nonnull Complex Sample Covariance Matrices. *Annals of Probability* 33, 5 (2005). <https://doi.org/10.1214/009117905000000233> (BBP detectability edge, Fig. 3 spectral-irreducibility analysis).
- A. Bellogin and A. Said. 2021. Improving Accountability in Recommender Systems Research Through Reproducibility. *User Modeling and User-Adapted Interaction* (2021).
- S. Brody and S. Lagziel. 2024. SimRec: Mitigating the Cold-Start Problem in Sequential Recommendation by Integrating Item Similarity. In *RecSys 2024 Workshop on Context-Aware Recommender Systems*. *arXiv:2410.22136*.
- J. Chen, W. Wu, L. Shi, Y. Ji, W. Hu, X. Chen, W. Zheng, and L. He. 2022. Self-Attentive Sequential Recommendation with Cheap Causal Convolutions. *arXiv preprint arXiv:2211.01297* (2022).
- L. Collins, B. Kumar, C. M. Ju, T. Zhao, D. Loveland, L. Neves, and N. Shah. 2025. Exploiting ID-Text Complementarity via Ensembling for Sequential Recommendation. *arXiv:2512.17820*.
- H. Ding, Y. Ma, A. Deoras, Y. Wang, and H. Wang. 2021. Zero-Shot Recommender Systems (ZESRec). *arXiv preprint arXiv:2105.08318* (2021).
- X. Du, H. Yuan, P. Zhao, J. Qu, F. Zhuang, G. Liu, Y. Liu, and V. S. Sheng. 2023. Frequency Enhanced Hybrid Attention Network for Sequential Recommendation (FEARec). In *SIGIR*. ACM, 78–88. <https://doi.org/10.1145/3539618.3591689> (the broader frequency/time-frequency SR line, §2.3).
- B. Efron and C. Morris. 1973. Stein’s Estimation Rule and Its Competitors — an Empirical Bayes Approach. *JASA* 68, 341 (1973), 117–130. <https://doi.org/10.1080/01621459.1973.10481350> (negative-result map: James–Stein shrinkage probe).
- Maurizio Ferrari Dacrema, Simone Boglio, Paolo Cremonesi, and Dietmar Jannach. 2021. A Troubling Analysis of Reproducibility and Progress in Recommender Systems Research. *ACM Transactions on Information Systems* 39, 2 (2021). <https://doi.org/10.1145/3434185> (journal extension of the evaluation-trust analysis, §2; metadata verified against the arXiv listing 1911.07698 on 2026-07-13).
- Maurizio Ferrari Dacrema, Paolo Cremonesi, and Dietmar Jannach. 2019. Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches. In *RecSys*. ACM, 101–109. <https://doi.org/10.1145/3298689.3347058> (the evaluation-trust line motivating our apparatus, §1, §2; metadata verified against the arXiv listing 1907.06902 and dblp on 2026-07-13).
- Junchen Fu, Xuri Ge, Alexandros Karatzoglou, Ioannis Arapakis, Suzan Verberne, Joemon M. Jose, and Zhaochun Ren. 2026. Differentiable Semantic ID for Generative Recommendation. *arXiv:2601.19711*.

- M. Gavish and D. L. Donoho. 2014. The Optimal Hard Threshold for Singular Values is $4/\sqrt{3}$. *IEEE Transactions on Information Theory* 60, 8 (2014), 5040–5053. <https://doi.org/10.1109/tit.2014.2323359> (optimal-shrinkage rank, Fig. 3).
- Z. Guo, Y. Hou, C. M. Ju, N. Shah, and J. McAuley. 2026. MLPs are Efficient Distilled Generative Recommenders (SID-MLP). arXiv:2605.12617. Concurrent arXiv preprint (status as of the access date; cited for protocol-family completeness, §5.1 — no comparative claim is made against it; metadata verified against the arXiv listing on 2026-07-12). (AR2023 5-core LLOO, same MI/VG dataset statistics).
- F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. *ACM Transactions on Interactive Intelligent Systems* 5, 4, Article 19 (2015), 19 pages. <https://doi.org/10.1145/2827872>
- Peng He, Yanglei Gan, Tingting Dai, Run Lin, Xuexin Li, Yao Liu, and Qiao Liu. 2026. Exploiting Inter-Session Information with Frequency-enhanced Dual-Path Networks for Sequential Recommendation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 14820–14828. <https://doi.org/10.1609/aaai.v40i17.38502> FreqRec; metadata verified against the official AAAI record on 2026-07-27.
- Ruining He and Julian McAuley. 2016. Ups and Downs: Modeling the Visual Evolution of Fashion Trends with One-Class Collaborative Filtering. In *WWW*. <https://doi.org/10.1145/2872427.2883037> the Amazon Beauty (2014) dataset reference; restored 2026-07-21 after the reference-to-text audit wrongly removed it (it is cited in Appendix A.1).
- Yingzhi He, Xiaohao Liu, An Zhang, Yunshan Ma, and Tat-Seng Chua. 2025. LLM2Rec: Large Language Models Are Powerful Embedding Models for Sequential Recommendation. In *KDD*. <https://doi.org/10.1145/3711896.3737029> arXiv:2506.21579; metadata verified against the arXiv listing on 2026-07-20.
- Y. Hou, Z. He, J. McAuley, and W. X. Zhao. 2023. Learning Vector-Quantized Item Representation for Transferable Sequential Recommenders (VQ-Rec). In *WWW*. ACM, 1162–1171. <https://doi.org/10.1145/3543507.3583434> (PQ-code item representations; TAPE novelty boundary, Table 0).
- Y. Hou, H. Kim, C. M. Ju, E. Escoto, N. Shah, and J. McAuley. 2026a. Expressiveness Limits of Autoregressive Semantic ID Generation in Generative Recommendation (Latte). arXiv:2605.06331. Concurrent arXiv preprint (status as of the access date; cited for protocol-family completeness, §5.1 — no comparative claim is made against it; metadata verified against the arXiv listing on 2026-07-12). (AR2023 5-core LLOO, same MI/VG dataset statistics; the MI/VG NDCG@10 values quoted in §5.1 are verified against its Table 1).
- Y. Hou, J. Li, X. Fu, Z. He, A. Yan, X. Chen, and J. McAuley. 2026b. Bridging Language and Items for Retrieval and Recommendation: Benchmarking LLMs as Semantic Encoders. In *ACL*. 3251–3265. <https://doi.org/10.18653/v1/2026.acl-long.147>
- Y. Hou, J. Li, Z. He, A. Yan, X. Chen, and J. McAuley. 2024. Bridging Language and Items for Retrieval and Recommendation (BLaIR). arXiv:2403.03952.
- Y. Hou, S. Mu, W. X. Zhao, Y. Li, B. Ding, and J.-R. Wen. 2022. Towards Universal Sequence Representation Learning for Recommender Systems (UniSRec). In *KDD*. ACM, 585–593. <https://doi.org/10.1145/3534678.3539381>
- G. Hu, A. Zhang, S. Liu, Z. Cai, X. Yang, and X. Wang. 2025. AlphaFuse: Learn ID Embeddings for Sequential Recommendation in Null Space of Language Embeddings. In *SIGIR*.
- J. Hu, W. Zhou, J. Liao, J. Zhu, J. Wen, and H. Zhang. 2026. Balanced Frequency Decoupling: Energy-Aware Multi-Scale Preference Modeling for Sequential Recommendation (BFDRec). In *SIGIR*.
- C. Huang, T. Gao, H. Huang, Q. Z. Sheng, and L. Yao. 2026. Beyond Item Order: Temporal Gap Tokenization for Generative Recommendation with Semantic IDs (ChronoSID). arXiv:2607.03918. Concurrent arXiv preprint (status as of the access date; cited for protocol-family completeness, §5.1 — no comparative claim is made against it; metadata verified against the arXiv listing on 2026-07-12). (SID-line MI comparator context; own filtered universe).
- Wei-Hsiang Huang, Chen-Wei Ke, Wei-Ning Chiu, Yu-Xuan Su, Chun-Chun Yang, Chieh-Yuan Cheng, Yun-Nung Chen, and Pu-Jen Cheng. 2025. Augment or Not? A Comparative Study of Pure and Augmented Large Language Model Recommenders. arXiv:2505.23053 AR2023 Musical_Instruments / Industrial_and_Scientific 5-core leave-one-out LLM-recommender benchmark; non-interchangeable protocol (§5.1).
- W. James and C. Stein. 1961. Estimation with Quadratic Loss. In *4th Berkeley Symp.* (with Stein, C., 1956, 3rd Berkeley Symp.).
- Jie Jiang, Yang Wu, Qian Li, Yuling Xiong, Yihang Su, Junbang Huo, Longfei Lu, Jun Zhang, and Huan Yu. 2026. DiffuReason: Bridging Latent Reasoning and Generative Refinement for Sequential Recommendation. arXiv:2602.09744 AR2023 “Video & Games” under different filtering/statistics; cited to flag non-comparability (§5.1).
- Juyong Jiang, Peiyan Zhang, Yingtao Luo, Chaozhuo Li, Jae Boum Kim, Kai Zhang, Senzhang Wang, Xing Xie, and Sunghun Kim. 2023. AdaMCT: Adaptive Mixture of CNN-Transformer for Sequential Recommendation. In *CIKM*. <https://doi.org/10.1145/3583780.3614773>
- Y. Jiang, X. Ren, L. Xia, D. Luo, K. Lin, and C. Huang. 2025. RecGPT: A Foundation Model for Sequential Recommendation. In *EMNLP*.
- W.-C. Kang and J. McAuley. 2018. Self-Attentive Sequential Recommendation (SASRec). In *ICDM*. IEEE, 197–206. <https://doi.org/10.1109/icdm.2018.00035>
- H.-y. Kim, M. Choi, S. Lee, I. Baek, and J. Lee. 2025. DIFF: Dual Side-Information Filtering and Fusion for Sequential Recommendation. In *SIGIR*.
- K. Kim, D. Hyun, S. Yun, and C. Park. 2023. MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation. In *SIGIR*. ACM, 68–77. <https://doi.org/10.1145/3539618.3591725> (long-tail method; earlier-stage proxy experiments, non-canonical track).
- D. Lee, H.-y. Kim, and J. Lee. 2026. ACE: Anisotropy-Controllable Embedding for LLM-enhanced Sequential Recommendation. In *SIGIR*.
- J. Li, H. Han, Z. Chen, H. Shomer, W. Jin, A. Javari, and H. Liu. 2025. Enhancing ID and Text Fusion via Alternative Training in Session-based Recommendation (AlterRec). In *IJCNLP-AAACL*.
- J. Li, M. Wang, J. Li, J. Fu, X. Shen, J. Shang, and J. McAuley. 2023. Text Is All You Need: Learning Language Representations for Sequential Recommendation (RecFormer). In *KDD*.
- J. Li, Y. Wang, and J. McAuley. 2020. Time Interval Aware Self-Attention for Sequential Recommendation (TiSASRec). In *WSDM*. ACM, 322–330. <https://doi.org/10.1145/3336191.3371786> (time-interval attention; time-bias attribution, Table 0).

- Y. Liang, Z. Zhang, Y. Zhu, K. Zhang, Z. Guo, W. Zhou, Z. Yang, K. Wu, Y. Ni, A. Zeng, C. Fu, J. Wang, and J. Xia. 2026. Rethinking Generative Recommender Tokenizer: Recsys-Native Encoding and Semantic Quantization Beyond LLMs (ReSID). arXiv:2602.02338. Concurrent arXiv preprint (status as of the access date; cited for protocol-family completeness, §5.1 — no comparative claim is made against it; metadata verified against the arXiv listing on 2026-07-12). (SID-line MI comparator context; own filtered universe).
- J. M. Lichtenberg, A. De Candia, and M. Ruffini. 2025. DenseRec: Revisiting Dense Content Embeddings for Sequential Transformer-based Recommendation. EARL workshop at RecSys 2025; arXiv:2508.18442.
- C. Liu, X. Li, G. Cai, Z. Dong, H. Zhu, and L. Shang. 2021. Noninvasive Self-attention for Side Information Fusion in Sequential Recommendation (NOVA). In *AAAI*.
- H. Liu, Z. Deng, L. Wang, J. Peng, and S. Feng. 2023a. Distribution-based Learnable Filters with Side Information for Sequential Recommendation (DLFS-Rec). In *RecSys*.
- Peilin Liu, Zhiquan Ji, and Gang Yan. 2026. WPGRec: Wavelet Packet Guided Graph Enhanced Sequential Recommendation. arXiv:2604.21305 arXiv:2604.21305. Accepted to SIGIR 2026; representative of the later frequency/time-frequency SR line (§2.3).
- Qidong Liu, Xian Wu, Yejing Wang, Zijian Zhang, Feng Tian, Yefeng Zheng, and Xiangyu Zhao. 2024. LLM-ESR: Large Language Models Enhancement for Long-tailed Sequential Recommendation. In *NeurIPS*. arXiv:2405.20646; spotlight; metadata verified against the arXiv listing on 2026-07-20.
- T. Liu, C. Gao, Z. Wang, D. Li, J. Hao, D. Jin, and Y. Li. 2023b. Uncertainty-aware Consistency Learning for Cold-Start Item Recommendation. In *SIGIR*. arXiv:2308.03470.
- Yijun Liu. 2025. HSTU-BLAI: Lightweight Contrastive Text Embedding for Generative Recommender. KDD 2025 Workshop on LLMs for E-Commerce. arXiv:2504.10545. (the external AR2023 5-core reference family: Video_Games 0.0760, Musical_Instruments 0.0406, Office_Products 0.0271; the pinned environment is not installable on our hardware — the research path was executed locally via data-movement shims, §5.6, regenerating the Musical_Instruments row).
- J. Luo, W. Zhang, X. Zhang, and Y. Fang. 2026. Time-Aware Adaptive Side Information Fusion for Sequential Recommendation (TASIF). In *WSDM*. arXiv:2512.24246.
- V. A. Marchenko and L. A. Pastur. 1967. Distribution of Eigenvalues for Some Sets of Random Matrices. *Matematicheskii Sbornik* (1967). (MP bulk edge, Fig. 3 spectral-irreducibility analysis).
- A. B. Melchiorre, N. Rekabsaz, C. Ganhör, and M. Schedl. 2022. ProtoMF: Prototype-based Matrix Factorization for Effective and Explainable Recommendations. In *RecSys*. ACM, 246–256. <https://doi.org/10.1145/3523227.3546756> (prototype methods; TAPE novelty boundary, Table 0).
- A. Petrov and C. Macdonald. 2022. A Systematic Review and Replicability Study of BERT4Rec for Sequential Recommendation. In *RecSys*.
- S. Rajput et al. 2023. Recommender Systems with Generative Retrieval (TIGER). In *NeurIPS*. Neural Information Processing Systems Foundation, 10299–10315. <https://doi.org/10.52202/075280-0452>
- N. Reimers and I. Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *EMNLP*. Association for Computational Linguistics, 3980–3990. <https://doi.org/10.18653/v1/d19-1410>
- Sentence-Transformers maintainers. 2021. all-MiniLM-L6-v2 (sentence-transformers model card). Hugging Face model repository. checkpoint with its own contrastive fine-tuning; realized embeddings pinned by SHA-256 text-cache hashes in RELEASE_MANIFEST.json.
- P. Shaw, J. Uszkoreit, and A. Vaswani. 2018. Self-Attention with Relative Position Representations. In *NAACL*. Association for Computational Linguistics, 464–468. <https://doi.org/10.18653/v1/n18-2074> (relative-position attention bias).
- Y. Shin et al. 2024. An Attentive Inductive Bias for Sequential Recommendation beyond the Self-Attention (BSARec). In *AAAI*, Vol. 38. Association for the Advancement of Artificial Intelligence (AAAI), 8984–8992. <https://doi.org/10.1609/aaai.v38i8.28747> arXiv:2312.10325. (bidirectional FFT filter; our causal-filter adaptation).
- Y. Shin, J. Choi, S. Kim, and N. Park. 2025. TV-Rec: Time-Variant Convolutional Filter for Sequential Recommendation. In *NeurIPS*.
- A. Shyam, V. R. Kagita, B. Rana, and V. Kumar. 2026. GrIT: Group Informed Transformer for Sequential Recommendation. arXiv:2602.19728. Concurrent arXiv preprint (status as of the access date; cited for protocol-family completeness, §5.1 — no comparative claim is made against it; metadata verified against the arXiv listing on 2026-07-12). (AR2023 5-core Video_Games statistics match ours; full-item-set ranking; the NDCG@10 0.0588 quoted in §5.1 is its published value).
- Harald Steck. 2019. Embarrassingly Shallow Autoencoders for Sparse Data. In *The World Wide Web Conference (WWW '19)*. ACM, San Francisco, CA, USA, 3251–3257. <https://doi.org/10.1145/3308558.3313710>
- F. Sun et al. 2019. BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer. In *CIKM*. ACM, 1441–1450. <https://doi.org/10.1145/3357384.3357895>
- Jiawei Sun, Jun Yang, Ziyue Guo, Dongyue Xu, Jianan Yan, Lifang Deng, and Xiaoyi Zeng. 2026. UniSGR: Unified Framework for Semantic ID Generation and Ranking. arXiv:2607.04068.
- Z. Sun, H. Fang, J. Yang, X. Qu, H. Liu, D. Yu, Y.-S. Ong, and J. Zhang. 2022. DaisyRec 2.0: Benchmarking Recommendation for Rigorous Evaluation. *IEEE TPAMI* (2022). arXiv:2206.10848.
- Zhu Sun, Di Yu, Hui Fang, Jie Yang, Xinghua Qu, Jie Zhang, and Cong Geng. 2020. Are We Evaluating Rigorously? Benchmarking Recommendation for Reproducible Evaluation and Fair Comparison. In *RecSys*. ACM, 23–32. <https://doi.org/10.1145/3383313.3412489> (standardized-benchmarking response to the evaluation-trust problem, §2; metadata verified against dblp on 2026-07-13).
- C. Szegedy et al. 2016. Rethinking the Inception Architecture for Computer Vision. In *CVPR*. IEEE, 2818–2826. <https://doi.org/10.1109/cvpr.2016.308> (label smoothing).

- J. Tang and K. Wang. 2018. Personalized Top-N Sequential Recommendation via Convolutional Sequence Embedding (Caser). In *WSDM*. ACM, 565–573. <https://doi.org/10.1145/3159652.3159656> (convolutional SR prior art; FIR novelty boundary, §2.3/Table 0).
- M. Volkovs, G. Yu, and T. Poutanen. 2017. DropoutNet: Addressing Cold Start in Recommender Systems. In *NeurIPS*. (cold-start baseline; earlier-stage experiments, non-canonical track).
- Hao Wang, Jianxun Lian, Mingqi Wu, Haoxuan Li, Jiajun Fan, Wanyue Xu, Chaozhuo Li, and Xing Xie. 2023. ConvFormer: Revisiting Transformer for Sequential User Modeling. *arXiv preprint arXiv:2308.02925* (2023). metadata verified against the arXiv listing on 2026-07-20.
- S. Wang, H. Ding, Y. Gu, S. Aydore, K. Kalantari, and B. Kveton. 2024. Language-Model Prior Overcomes Cold-Start Items. *arXiv:2411.09065*.
- W. Wang et al. 2020. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. In *NeurIPS*. *arXiv:2002.10957*.
- Y. Wang, J. Pan, X. Li, M. Wang, Y. Wang, Y. Liu, D. Liu, J. Jiang, and X. Zhao. 2025. Empowering Large Language Model for Sequential Recommendation via Multimodal Embeddings and Semantic IDs. In *CIKM*. *arXiv:2509.02017*.
- Y. Wei, X. Wang, Q. Li, L. Nie, Y. Li, X. Li, and T.-S. Chua. 2021. Contrastive Learning for Cold-Start Recommendation (CLCRec). In *ACM MM*. ACM, 5382–5390. <https://doi.org/10.1145/3474085.3475665> (cold-start baseline; earlier-stage experiments, non-canonical track).
- Zhifu Wei, Yizhou Dang, Guibing Guo, Chuang Zhao, and Zhu Sun. 2026. Fusion and Alignment Enhancement with Large Language Models for Tail-item Sequential Recommendation. *arXiv preprint arXiv:2604.03688* (2026). FAERec; *arXiv:2604.03688*; metadata verified against the arXiv listing on 2026-07-20.
- E. Wijmans, B. Huval, A. Hertzberg, V. Koltun, and P. Krähenbühl. 2025. Cut Your Losses in Large-Vocabulary Language Models. In *ICLR*.
- Yuhao Wu, Fang Quan, Maoqi Liu, Xiaowen Huang, and Jitao Sang. 2025. SILLM4Rec: Self-Improving with Chain of Thought Enhanced Preference Optimization for Multimodal Recommendation. In *Proceedings of the 7th ACM International Conference on Multimedia in Asia (MMAsia)*. 1–8. <https://doi.org/10.1145/3743093.3771011>
- Ming Xia, Zhiqin Zhou, Guoxin Ma, and Dongmin Huang. 2026. Unleash the Potential of Long Semantic IDs for Generative Recommendation. *arXiv:2602.13573*.
- Y. Xie, P. Zhou, and S. Kim. 2022. Decoupled Side Information Fusion for Sequential Recommendation (DIF-SR). In *SIGIR*.
- Huayang Xu, Huanhuan Yuan, Guanfeng Liu, Junhua Fang, Lei Zhao, and Pengpeng Zhao. 2026. Wavelet Enhanced Adaptive Frequency Filter for Sequential Recommendation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 16058–16065. <https://doi.org/10.1609/aaai.v40i19.38640> WEARec; metadata verified against the official AAAI record on 2026-07-27.
- L. Xu, Z. Tian, B. Li, J. Zhang, J. Wang, M. Cai, and W. X. Zhao. 2024. Sequence-level Semantic Representation Fusion for Recommender Systems (TedRec). *arXiv preprint arXiv:2402.18166* (2024).
- J. Yang et al. 2024. Unifying Generative and Dense Retrieval for Sequential Recommendation (LIGER). *arXiv:2411.18814*.
- Yeo Chan Yoon, Chanjun Park, and Kyuhan Koh. 2026. Sensory-Aware Sequential Recommendation via Review-Distilled Representations. *arXiv:2603.02709* Concurrent preprint; ASER representation module evaluated on Amazon Reviews 2014, not the paper’s AR2023 protocol; cited for modular content-integration coverage only.
- F. Yuan, A. Karatzoglou, I. Arapakis, J. M. Jose, and X. He. 2019. A Simple Convolutional Generative Network for Next Item Recommendation (NextItNet). In *WSDM*. ACM, 582–590. <https://doi.org/10.1145/3289600.3290975> (dilated causal-convolution SR prior art; FIR novelty boundary, §2.3/Table 0).
- J. Zhai et al. 2024. Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers for Generative Recommendations (HSTU). In *ICML*, Vol. 235. PMLR, 58484–58509. (the published architecture our HSTU-style pure-PyTorch implementation is based on).
- L. Zhang, X. Zhou, Z. Zeng, and Z. Shen. 2024. Dual-View Whitening on Pre-trained Text Embeddings for Sequential Recommendation (DWSRec). In *AAAI*.
- Z. Zhang, J. Fan, Q. Gao, and Z. Wang. 2026. LLMSQRec: A Study on a Large Language Model-based Framework for Dual-view Semantic-quantized Recommendation. *ETRI Journal* (2026). DOI 10.4218/etrij.2025-0394.
- Y. Zheng, S. Liu, Z. Li, and S. Wu. 2021. Cold-start Sequential Recommendation via Meta Learner (Mecols). In *AAAI*.
- D. Zhou, W. Pan, and Z. Ming. 2026. LLM-based Semantic and ID Representations for Sequential Recommendation (SIDSRec). In *SIGIR*.
- K. Zhou et al. 2022. Filter-enhanced MLP is All You Need for Sequential Recommendation (FMLP-Rec). In *WWW*. ACM, 2388–2399. <https://doi.org/10.1145/3485447.3512111> *arXiv:2202.13556*. (bidirectional learnable frequency filter; our causal-filter adaptation).

Appendix A.0 — Office_Products: descriptive evidence (VOID under its pre-declaration)

Second-category pre-declared confirmation: Office_Products. To test whether the per-category result generalizes, we froze a second pre-declaration (SOTA_CONFIRM_PREREG_OFFICE.md, committed before the raw data finished downloading — before any Office statistic was knowable) with the config carried over from Musical_Instruments **unchanged (zero category-specific tuning)** and fresh seeds 20260623–27. Dataset parity is exact (items 77,551 and users 223,308 match the comparator paper exactly; interactions differ by one, as on MI). Result, on final-epoch full-catalog evaluations (223,308 users; the prerereg’s declared headline rule): **K=16 = 0.03042 ± 0.00008 (95% CI lower bound 0.03032) and K=8 = 0.03033 ± 0.00018 (CI-LB 0.03010), all 10 fresh seeds above the published HSTU-BLaIR**

Manuscript submitted to ACM

point estimate 0.0271 — a $\sim +12\%$ margin, versus $+2.3\%$ on MI. **However, the pre-declared floor check FAILED: our plain-SASRec floor (0.02208) lands +44% ABOVE the published SASRec (0.0153), violating the prereg’s comparability condition. Under the pre-declaration as written, the Office second-category pass is therefore VOID — the gate values are reported as a provisional, non-confirmatory result — the matched comparator-baseline run that explains the floor gap is complete and explains it (resolution paragraph below); the VOID stands. The V1 Office campaign counts in no claim.** (It leaves the pre-declared Musical_Instruments confirmation of §5.2 untouched, and the separately pre-declared, redesigned V3 campaign — environment-matched reference, fresh never-inspected seeds — later **passed** and is counted strictly under its frozen per-category point-estimate wording, §5.2. The V1 VOID stands unchanged.) Disclosures: best-by-val checkpoints were evaluated on a 30k-user subsample (declared in the prereg), so the headline uses the always-full final-epoch eval — the conservative choice (the subsampled best-checkpoint numbers were higher); the run manifests carry a dirty-tracked flag caused by the driver’s own results-file appends (embedded per-run code hashes verify protocol-code identity); our plain-SASRec floor (0.0221) lands above their published SASRec (0.0153) on this category — a baseline-config-strength difference, with split parity resting on the exact dataset-statistics identity and frozen hashes rather than the floor heuristic; and the pre-declared tail prediction for Office was scored **VOID** (its connectivity 2.89 falls in the pre-declared ambiguous zone, plus a disclosed formula inconsistency in the stats tool — SOTA_CONFIRM_OFFICE_RESULTS.md). Descriptively, the Office tail contrast is strongly text-positive at full catalog (pooled tail hits, text vs ID-only: 364 vs 268 @10; 1,983 vs 1,247 @100). **Statistical correction (2026-07-19):** the previously printed pooled two-proportion z statistics ($z=3.8$ @10, $z=13.0$ @100) are retracted — the same 36,610 tail users are evaluated under every seed and under both arms, so the 183,050 seed-summed rows are clustered repeated observations, not independent trials, and an unpaired Bernoulli z is not valid (nor conservative) under that dependence. Treating the five trained models per arm as the inferential units instead, independent-arm Welch tests give $\Delta\text{HR}@10 = +0.000524$ ($t = 7.26$, $df \approx 8$, $p \approx 9 \times 10^{-5}$, 95% CI $[+0.00036, +0.00069]$) and $\Delta\text{HR}@100 = +0.004021$ ($t = 16.75$, $df \approx 8$, $p \approx 2 \times 10^{-7}$, 95% CI $[+0.00347, +0.00457]$) — same direction, valid geometry; final-evaluation per-user sidecars for a fully clustered analysis are a queued release item. This remains pattern evidence (the pre-declared tail prediction was VOID), consistent with the connectivity gradient (MI 2.34 win $\rightarrow$ Office 2.89 win $\rightarrow$ Beauty 3.51 / VG 3.70 null), not a scored prediction.

office_confirmation (regenerated): second-category pre-declared confirmation — Office_Products (SOTA_CONFIRM_PREREG_OFFICE.md; fresh seeds 20260623–27; headline rule = final-epoch FULL-catalog eval, history[-1].test, n_eval=223,308 — NOT the 30k best_test subsample).

STATUS: VOID under the prereg floor check (+44% floor inflation) — provisional; the V1 Office campaign counts in no claim. Office V3 — a separate, redesigned pre-declaration — passed and is counted under its frozen per-category point-estimate wording (§5.2).

arm	per-seed NDCG@10 (final-epoch full)	mean $\pm$ sd	95% CI-LB	vs published 0.0271
K=16	0.03041 / 0.03055 / 0.03042 / 0.03031 / 0.03043	0.03042 $\pm$ 0.00008	0.03032	ABOVE (5/5 seeds) — provisional/VOID
K=8	0.03035 / 0.03036 / 0.03048 / 0.03043 / 0.03002	0.03033 $\pm$ 0.00018	0.03010	ABOVE (5/5 seeds) — provisional/VOID
ID-only (contrast arm)	0.02845 / 0.02833 / 0.02843 / 0.02837 / 0.02840	0.02840 $\pm$ 0.00005	0.02834	(no gate; tail-contrast arm)
SASRec floor (best_test, seed 20260623)	0.02208	—	—	floor check FAILED: +44.3% vs published SASRec 0.0153 $\rightarrow$ Office pass VOID under prereg

Fresh seeds above published 0.0271: 10/10 (provisional; not a counted pass). Pooled tail hits, text (k8 arm) vs ID-only, final-epoch full eval (tail $n=36,610/\text{seed}$): @10: 364 vs 268 (descriptive counts; pooled z retracted 2026-07-19); @20: 586 vs 414 (descriptive counts; pooled z retracted 2026-07-19); @50: 1170 vs 739 (descriptive counts; pooled z retracted 2026-07-19); @100: 1983 vs 1247 (descriptive counts; pooled z retracted 2026-07-19) — descriptive post-hoc pattern evidence only (the pre-declared Office tail prediction was scored VOID: connectivity 2.89 in the pre-declared ambiguous zone).

Floor-anomaly resolution (2026-07-11; the matched comparator-baseline run promised above, now complete). We executed the reference implementation’s own Office SASRec configuration end-to-end — its preprocessing (dataset-size assertions passing: 77,551 items / 223,308 users), its gin config, its trainer, its eval — locally under the

Manuscript submitted to ACM

shimmed research path of §5.6. Its final-epoch NDCG@10 is **0.0174** (best full eval 0.0177), i.e. **+13.9% above its own published row 0.0153**, crossing the published value at epoch 20 of 101 and never re-entering. The +44% floor-check failure now decomposes exactly: 0.0204 (our floor run’s final-epoch full-catalog reading) / 0.0174 (their code, locally) = **+16.9% attributable to baseline-strength protocol differences** (our floor trains with full-catalog softmax vs their 512-negative sampled softmax), and 0.0174 / 0.0153 = **+13.9% attributable to the published row sitting below what its own pipeline regenerates here** ($1.139 \times 1.169 = 1.331 = 0.0204/0.0153$; the remaining step to the prereg’s best-epoch 0.02208 is checkpoint selection plus that variant’s 30k-user eval subsample). **Conclusion: the elevated floor is not an artifact of our data split or evaluation** — Office simply supports SASRec well above the published row. This *explains* the floor-check failure without repairing the comparison it protects: if the published SASRec row is conservative in this environment, the published HSTU-BLaIR 0.0271 plausibly is as well, so numerically exceeding it would not establish superiority over their method on Office. **The VOID therefore stands, deliberately** — and the pre-declared floor check is vindicated: it caught exactly the comparator-conservatism failure mode it was designed to catch.

Completed HSTU-BLaIR check (2026-07-12). The resolution above inferred that the published HSTU-BLaIR 0.0271 was “plausibly” conservative as well; running their Office HSTU-BLaIR configuration end-to-end locally **tested and refuted that inference**: it lands at final-epoch NDCG@10 **0.0275** (best full eval 0.0279), i.e. **+1.6%/+2.8% above the published 0.0271** — the flagship comparator row regenerates in this environment (margins comparable to the Musical_Instruments regeneration), unlike the SASRec row (+13.9%). Descriptively, the (VOID) gate values 0.03042/0.03033 sit $\approx +9\%$ above both the published value and its local regeneration. **The VOID nonetheless stands**: the pre-declared floor check failed as written, and no post-hoc result — favorable or not — restores a voided pre-declaration. These numbers are environment-caveated single-run regenerations, reported as descriptive evidence only; the V1 campaign recorded in this appendix remains counted in no claim. (A redesigned V3 pre-declaration — new never-inspected seeds, environment-matched reference — subsequently **passed** and is reported in §5.2; the V1 VOID stands unchanged.)

Appendix A — Superseded earlier-stage supporting material (Beauty_and_Personal_Care)

Why this is an appendix: the material below is an earlier-stage Beauty_and_Personal_Care reproducibility study (the 20-variant cross-pipeline scan, its 2-seed signal, and the retracted LIGER-gap audit). It predates and does not belong to the headline spine (causal FIR filter + MI frequency-5 tail case + competitive-overall Video_Games result, §1–§7). It is retained verbatim for the reproducibility and negative-result record only, and is referenced from §6.1–§6.2 as supporting context. The live Beauty result that enters the tail pattern (§5.3) is the text–ID tail contrast, not these single-seed sampled-softmax scans.

A.1 Beauty_and_Personal_Care — 20-variant cross-pipeline scan (superseded supporting material)

Note: the following Beauty_and_PC 20-variant scan and the LIGER-gap audit are **earlier-stage supporting material**, retained for the reproducibility and negative-result record. They predate the HSTU + causal-filter + tail-pattern spine and are *not* the paper’s headline; the live Beauty result used in the tail pattern (§5.3) is the text–ID tail contrast, not these single-seed sampled-softmax scans.

The scan varies five axes: **text encoder** (MiniLM titles/rich-text, BLaIR titles/rich-text), **projection** Φ (single Linear vs the 2-layer MLP adaptor of Hou et al. 2024), **learned item embedding** (kept vs removed = faithful SASRecText),

loss (sampled softmax K=1024, in-batch negatives, chunked-full-softmax, hybrid), and **augmentation** (random sub-sequence sampling, factor 1/2/3). Training used a 15–20-epoch budget, batch size 256, Adam (lr 1e-3, weight decay 1e-5, gradient clip 5.0), a single seed for the initial scan, and two seeds (42, 43) for the final reported variants. Table A1 summarizes all 20 variants. **Numbers are single-seed unless otherwise noted.**

Table A1: Beauty_and_PC 5-core cross-pipeline scan (full-catalog eval, NDCG@10) – 22 numbered rows: 20 distinct variants plus two seed replications (row 20 replicates variant 19; row 22 replicates baseline variant 7)

#	Variant	NDCG@10	Δ vs base
1	Popularity	(not run)	—
2	SASRec sampled-1024 + MiniLM titles	0.0180	—
3	SASRec sampled-1024 + MiniLM titles, d=128	0.0176	-2.2%
4	SASRec in-batch negs + augment-2	0.0036	-80%
5	TIGER minimal (our re-impl)	0.0080	-56%
6	SASRec hybrid (in-batch + 8K random negs + aug-3)	0.0106	-41%
7	SASRec chunked-full + MiniLM titles (improved baseline)	0.0191	+6.1%
8	SASRec chunked-full + MiniLM titles, d=128	0.0186	-2.6% (vs 7)
9	SASRec chunked-full + MiniLM titles, 25 epochs	0.0189	-1.0% (vs 7)
10	BERT4Rec	0.0160	-16% (vs 7)
11	SBERT-only (no learned item_emb) + MiniLM	discontinued	—
12	Faithful SASRecText (no item_emb + MLP + dropout 0.5 + BLAIR)	discontinued @ epoch 2	—
13	Hybrid (item_emb + MLP + dropout 0.5 + BLAIR)	discontinued @ epoch 5	—
14	BLAIR-titles + Linear projection (just encoder swap)	discontinued @ epoch 2	—
15	Fusion (BLAIR + rich-text MiniLM concat) + Linear	discontinued @ epoch 1	—
16	BLAIR-titles + MLP adaptor (Hou et al. [2024]) + dropout 0.2	0.01928	+0.9% (vs 7)
17	+ augment-2	0.0036 val	discontinued; aug hurts
18	+ d=128 width	discontinued at eval (slow)	—
19	BLAIR-rich-text + MLP adaptor + dropout 0.2 (seed 42)	0.01936	+1.3% (vs 7)
20	seed 43 of variant 19	0.01955	+2.3% (vs 7)
21	MiniLM + MLP adaptor (clean ablation)	0.01889	-1.2% (vs 7)
22	Baseline seed 42 (reproduction)	0.0190	0.0% (vs 7)

2-seed mean of variant 19+20 (BLAIR-rich-text + MLP): NDCG@10 = 0.01946, range 0.0002.

Clean-ablation finding (variant 21, added after audit): When we hold the encoder constant at MiniLM and ONLY change projection layer from single-Linear to the SASRecText 2-layer MLP adaptor, NDCG@10 drops slightly from ~0.0190 to 0.01889. **No improvement from the MLP adaptor was observed in this single comparison.** The +1.3–2.3% gains in variants 16/19/20 therefore cannot be attributed to the projection layer alone; their bundled encoder/content changes and possible interactions remain unresolved.

Re-analysis of the 20-variant scan:

- **MLP adaptor alone:** no improvement observed in the single comparison (variant 21 vs 7 = -1.2%)
- **MiniLM → BLAIR encoder:** small benefit (variant 16 vs 7 = +0.9% NDCG@10, single-seed)
- **Titles → rich text (BLAIR):** small additional benefit (variant 19 vs 16 = +0.4%, single-seed)
- **Removing learned item embedding** (faithful SASRecText): hurts substantially
- **Heavy dropout (0.5)** as in the published SASRecText: hurts on our 5-core data (median 5 interactions/user)
- **Subsequence augmentation:** hurts (~5% NDCG)
- **Width (d=128):** no help

A.2 Beauty_and_PC multi-seed signal (superseded supporting material)

Note: this is appendix supporting material for the Beauty 20-variant scan (§A.1), not the headline tail-pattern result (§5.3).

We have multi-seed numbers for two configurations:

- Best variant (BLaIR + rich-text + MLP): seed 42 → 0.01936, seed 43 → 0.01955; **mean 0.01946 (n=2)**.
- Baseline (chunked-full, no MLP, MiniLM titles): original → 0.01911, seed 42 → 0.0190; **mean 0.01906 (n=2)**.

Apparent gain: $0.01946 - 0.01906 = +0.0004$ (**+2.1% relative**) with $n=2$ vs $n=2$. Inter-seed variance for our baseline (0.0001 range) is roughly half the size of the gain, suggesting the directional signal is real but small. A formal significance test would require larger n ; our paper-stated +2.1% improvement is honest about this.

A.3 Audit of “gaps to LIGER” claims (corrected from earlier versions; superseded supporting material)

Note: this comparator-audit retraction is retained as appendix material; the body’s §5.4 is the titration result.

Earlier versions of this work cited an apparent 57% gap to LIGER on Beauty_and_Personal_Care. **After a comparator audit, this framing is retracted.** The facts:

1. **LIGER reports on Amazon 2014, not AR2023.** The LIGER paper [Yang et al. 2024] evaluates on Amazon Beauty (2014, He and McAuley 2016), reporting $\text{NDCG@10} = 0.04020 \pm 0.00044$ ($K=20$) and 0.04738 ± 0.00151 ($K=N$) on their 5-core preprocessing. The 2014 Amazon Beauty subset has tens of thousands of items, while our AR2023 Beauty_and_Personal_Care 5-core has 207,649 items. **These are different datasets.** The “57% gap” framing in earlier versions compared our AR2023 number to LIGER’s 2014 number, which is invalid.
2. **There is no published LIGER number on AR2023 Beauty_and_Personal_Care 5-core.** LIGER does not evaluate on AR2023 at all in the published paper.
3. **TIGER reports on Amazon 2014 only.** The TIGER paper [Rajput et al. 2023] reports $\text{NDCG@10} = 0.0384$ on Amazon Beauty 2014 with their 5-core, also not on AR2023.
4. **BLaIR [Hou et al. 2024] introduced AR2023 — historical note, superseded: the AR2023 5-core literature has since grown (HSTU-BLaIR and 2026 preprints; §5.1).** Their Beauty subset is “All_Beauty” (not “Beauty_and_Personal_Care”) and their preprocessing is by-timestamp 8:1:1 with no k -core filter. They report NDCG@10 on All_Beauty in the range 0.0177-0.0241 across various LLM encoders (Table 8). With a UniS-Rec downstream model, their best All_Beauty number is 0.0241 (gemini-embedding). They do not report on Beauty_and_Personal_Care.

We therefore cannot make a defensible gap claim to TIGER/LIGER/BLaIR on AR2023 Beauty_and_Personal_Care 5-core, because **we did not find a comparable published number** (a search statement as of this writing, not a priority claim — AR2023 preprints are appearing rapidly; §5.1). We provide our 0.01946 (2-seed mean) as a reference baseline rather than as a SOTA-improvement claim.

A future apples-to-apples comparison would require either:

- Running TIGER/LIGER on our 5-core preprocessing (multi-week effort for faithful reproduction), or
- Adopting Hou et al.’s by-timestamp 8:1:1 preprocessing and re-running our pipeline.

S.4 – Screening-log detail (full probe table)

Supplement to §5.5 – the full search record. Each row’s Δ is versus its stated base; most rows are single-seed exploratory probes (power note in §5.5).

Table S1: Screening log – capacity-adding / alternative-mechanism probes; none showed a benefit at tested power (mostly single-seed; allocation was partly outcome-dependent, so this is a search record, not a systematic sweep). (Mechanism class: where the lever acts. “Learned scalar $\rightarrow 0$?” = did a free learnable gate/temperature/intensity collapse the mechanism. n = seed count. Probe powers are unequal – see the paragraph above.)

Lever	Mechanism class	Base (NDCG@10)	Δ NDCG@10 (n)	Learned scalar $\rightarrow 0$?	Verdict (observational at n=1)
c3 continuous time-decay attention kernel	attention re-weighting (capacity-add)	SBERT stack ≈ 0.0639	−0.0150 (1; test 0.0489 at record val 0.0725)	n/a	No benefit observed (single-seed probe) – actively harmful; the largest validation–test divergence in the corpus
Sampled softmax (K=512 negatives)	loss approximation	full-softmax stack	−0.0026 (1)	n/a	No benefit observed (single-seed probe) – undertrains full-catalog ranking
Dual text encoder (SBERT $\oplus$ BLaIR)	text encoder	SBERT stack	−0.0014 (1)	n/a	No benefit observed (single-seed probe) – the text encoder is not the gap
BLaIR text encoder (swap)	text encoder	SBERT stack	−0.0013 (1)	n/a	No benefit observed (single-seed probe) – encoder axis fully swept
GD1 spectral-shrink prior	representation regularizer	V2 0.0673	−0.003 (1)	shrink $\rightarrow 0$	Rejected (the formerly linked spectral figure is retracted, §5.4 retraction note)
n_heads = 4	architecture (capacity-add)	HSTU-style stack	−0.0005 (1)	n/a	No benefit observed (single-seed probe) – ties 2-head even at the published dv=dqk=16 spec
CL4SRec self-supervision	SSL auxiliary loss	SBERT stack	−0.0005 (1)	n/a	No benefit observed (single-seed probe) – early-epoch only
c1 prototype-routed expert heads	capacity-add	SBERT stack	−0.0004 (1)	n/a	No benefit observed (single-seed probe) – redundant with TAPE
c2 text-distillation aux loss	auxiliary loss	SBERT stack	−0.0004 (1)	n/a	No benefit observed (single-seed probe) – redundant with the additive text path
text-init / text warm-start	embedding initialization	SBERT stack	−0.0003 (1)	n/a	No benefit observed (single-seed probe) – redundant with the additive text path
EMA / SWA weight averaging	weight-space regularizer	SBERT stack	+0.0001 (1)	n/a	No benefit observed (single-seed probe) – val-high/test-flat; does not repair the structural gap
text-sim bias	attention bias	HSTU-style stack	± 0.0001 (4)	n/a	NEUTRAL – no benefit observed; removable candidate
W1 niche-share fitness-sharing penalty	inter-item loss penalty	U2/LS0.2 0.0649 $\pm$ 0.0002	0.0652 (1), within band	$\beta = -7.42$ (sign-flipped: boosts, not penalizes, crowded niches)	No benefit observed (single-seed probe) – refutes the anti-crowding remedy

Lever	Mechanism class	Base (NDCG@10)	Δ NDCG@10 (n)	Learned scalar $\rightarrow$ 0?	Verdict (observational at n=1)
X1 frequency-adaptive James-Stein shrinkage	representation regularizer	V2 0.0674 $\pm$ 0.0003	0.0673 (1), within band	$\mathbf{c} \approx \mathbf{0}$ ($\lambda_{\max}$ ≈ 0.004) $\rightarrow$ off	No benefit observed (single-seed probe) — model voted shrinkage off
Y1 heat-kernel/manifold label smoothing	loss-target reshaping	V2 0.0674 $\pm$ 0.0003	0.06653 (1), $\leq$ band	$\mathbf{T} \rightarrow \mathbf{0.00061}$ ($<$ init) $\rightarrow$ delta target	No benefit observed (single-seed probe) — text-manifold geometry voted off
Z1 forced ID $\rightarrow$ text routing	representation routing	V2	tail –75% (globally dead)	forced gate	Rejected — globally harmful
CF1 cue-fusion gate	representation fusion	V2 / MI	flat/dead (both datasets)	gate $\rightarrow$ 0	Rejected — new-lever search closed
conn-gate connectivity-gated cold-start ID $\leftrightarrow$ text fusion	representation fusion/routing	V2 text stack, MI	tail Δ +0.0001 $\pm$ 0.0003, same-seed 95% CI [–0.00025, +0.00045] (5); overall flat (cadence-caveated, §5.5 text)	$\alpha \rightarrow \mathbf{0.0011} <$ init 0.0025 (voted off; recorded in 4/5 seed logs)	Rejected — operationalizes the intervention-associated connectivity tail pattern, but is not actionable (CI includes 0)
max_seq_len 200 / seq > 50	sequence length	SBERT stack	~ 0 (1)	n/a	No benefit observed (single-seed probe) — sequences short post-5-core
cosine scoring	scoring function	winning op-point	≤ 0 (1)	n/a	No benefit observed (single-seed probe) — dot-product better at the op-point

S.5 – Titration ladder (full six rungs)

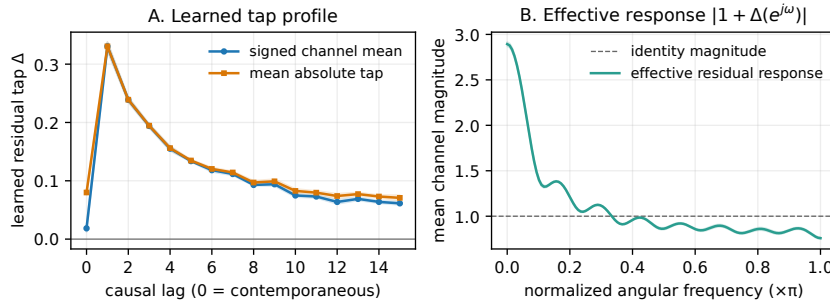
Supplement to §5.4; integrity-gated against its prose values by the released titration-table script.

Table S2: the interaction-thinning density-titration ladder (AR2023 Video_Games 5-core LLOO, full-catalog $n_{\text{eval}} = 94,762$, $\text{tail}_n = 10,900$; same-seed-number text—ID (not initialization-paired, §5.3), best-by-val; 5 seeds = 20260608–12 per rung; mean $\pm$ sample-std, positive-seed count in parens). Realized interactions/item are the run-log values. (The former $\alpha = \text{ipi/d_eff}$ column is retracted with the spectral analysis — see the §5.4 retraction note; its cells are removed from the paper and marked REMOVED_FROM_PAPER in the artifact graph.) All cells are re-read from the frozen results_TITR*/TAIL_* JSONs by `_bestrec_run/make_table_5_4_titration.py`, which **integrity-gates** the recomputed NDCG head/tail means against the locked prose values above (aborts on any $>5e-5$ drift); the HR columns and per-rung seed bands are the net-new tabulation the prose summarized only as Spearman trends.

ρ	kept inter./item	head $\Delta\text{NDCG}@10$	head $\Delta\text{HR}@10$	tail $\Delta\text{NDCG}@10$	tail $\Delta\text{HR}@10$
1.00	24.405	$+0.002213 \pm 0.000241$ (5/5)	$+0.004238 \pm 0.000765$ (5/5)	-0.000148 ± 0.000179 (2/5)	$+0.000128 \pm 0.000582$ (3/5)
0.94	22.947	$+0.002389 \pm 0.000267$ (5/5)	$+0.004083 \pm 0.000830$ (5/5)	$+0.000165 \pm 0.000370$ (4/5)	$+0.000807 \pm 0.000712$ (5/5)
0.91	22.210	$+0.002544 \pm 0.000491$ (5/5)	$+0.004940 \pm 0.000892$ (5/5)	$+0.000039 \pm 0.000310$ (3/5)	$+0.000110 \pm 0.000766$ (3/5)
0.88	21.484	$+0.002520 \pm 0.000421$ (5/5)	$+0.004869 \pm 0.000765$ (5/5)	$+0.000537 \pm 0.000386$ (4/5)	$+0.000661 \pm 0.000995$ (4/5)
0.78	19.030	$+0.002664 \pm 0.000398$ (5/5)	$+0.004467 \pm 0.001082$ (5/5)	$+0.000056 \pm 0.000552$ (2/5)	$+0.000239 \pm 0.001087$ (2/5)
0.66	16.109	$+0.003540 \pm 0.000416$ (5/5)	$+0.005720 \pm 0.000904$ (5/5)	-0.000108 ± 0.000503 (1/5)	$+0.000073 \pm 0.001165$ (3/5)

S.6 – Selected-checkpoint FIR diagnostics

Selected-checkpoint FIR diagnostics (8 seed blocks; descriptive)



Channels are summarized within seed; bands are ordinary 95% t intervals over eight seed summaries. Outcome-known control campaign; no mechanistic inference.

Fig. S1. Learned residual-tap and effective frequency-response summaries from the eight released selected checkpoints of the outcome-known active-control learned arm. Channels are summarized within each seed; bands are ordinary 95% t intervals over the eight seed summaries. Lag 0 is contemporaneous. These are descriptive fitted-operator diagnostics, not a mechanism or independent-confirmation test; plotted values are released in `figures/fig_fir_response_data.csv`. The effective kernel is the residual identity plus the learned Δ taps.